Online Appendix

to Havranek, T., Irsova, Z., Laslopova, L. & O. Zeynalova: *Publication and Attenuation Biases in Measuring Skill Substitution*, The Review of Economics and Statistics.

A The Elasticity Dataset

The elasticity of substitution between skilled and unskilled labor is usually defined as the change of the ratio in which these two factors are used in production divided by the change of the ratio of their marginal products. Under perfect competition, production factors are paid their marginal products and the elasticity can be written as

$$\sigma_{US} = \frac{\frac{d(L_U/L_S)}{L_U/L_S}}{\frac{d(w_S/w_U)}{w_S/w_U}} = -\frac{d\ log(L_U/L_S)}{d\ log(w_U/w_S)}, \tag{1}$$

where L_S and L_U denote skilled and unskilled labor; w_S and w_U denote their respective wage rates. Under a quasi-concave production function the elasticity of substitution attains any value from zero to infinity. If $\sigma = 0$, the two types of labor form perfect complements. Fixed proportions of the two inputs are needed to increase production; they cannot be substituted for each other. If $\sigma \in (0, 1)$, skilled and unskilled workers are gross complements: an increased supply of skilled workers increases the demand for unskilled workers. A unitary elasticity implies that relative quantity changes are exactly proportional to relative price changes. If $\sigma > 1$, skilled and unskilled workers form gross substitutes: unskilled workers can more easily work in positions intended for skilled workers (though with a lower productivity), and skilled workers can be tapped for a menial job. An increased supply of skilled workers decreases the demand for unskilled workers. Many researchers estimating the elasticity start with the following constant elasticity of substitution (CES) production function:

$$Y = [\alpha(aL_S)^\rho + (1-\alpha)(bL_U)^\rho]^{\frac{1}{\rho}}, \tag{2}$$

where skilled labor L_S and unskilled labor L_U are the sole factors of production, a and b are indices of factor-augmenting technology, and α is a technology parameter interpretable as indexing the "share of work" allocated to L_S. The elasticity can be derived from the parameter ρ as $\sigma = \frac{1}{1-\rho}$.

Whether researchers assume a one-level CES function or a nested one (also taking into account other inputs, such as capital), they typically employ the following steps. First, marginal products are obtained by taking derivatives of Y with respect to L_S and L_U. The assumption of competitive labor markets implies the equality of the wage ratio and the ratio of marginal products. Substituting $(\sigma - 1)/\sigma$ for ρ then leads to the definition of the skill premium $\frac{w_S}{w_U}$:

$$\frac{w_S}{w_U} = \frac{\alpha}{1-\alpha}\left(\frac{a}{b}\right)^{\frac{\sigma-1}{\sigma}}\left(\frac{L_S}{L_U}\right)^{-\frac{1}{\sigma}}. \tag{3}$$

Taking logarithms produces a specification that can be estimated:

$$\ln\left(\frac{w_S}{w_U}\right) = \ln\left(\frac{\alpha}{1-\alpha}\right) + \frac{\sigma-1}{\sigma}\ln\left(\frac{a}{b}\right) - \frac{1}{\sigma}\ln\left(\frac{L_S}{L_U}\right). \tag{4}$$

The main coefficient of interest, the negative inverse of the elasticity ($-1/\sigma$), can thus be interpreted as the causal effect of the relative supply of skilled labor on the wage premium to skills (in percentage terms). The term capturing skill-biased technical change (a/b), and thus demand for skills, is usually proxied by a time trend. Some authors estimate the regression in a reversed form (regressing relative skill supply on the skill premium), which gives them a direct estimate of the elasticity, not its inverse (see details in Appendix B). Details on various specification and estimation techniques employed in the literature are available in Appendix D. In data collection and reporting we follow the guidelines compiled by the Meta-Analysis in Economics Research Network (Havranek *et al.*, 2020).

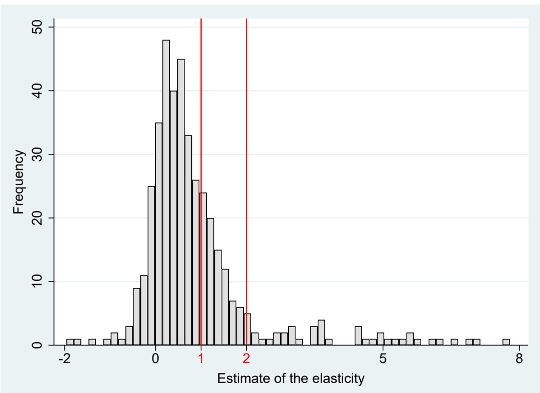
Table A1: The studies used in the meta-analysis (reporting both direct and inverse estimates)

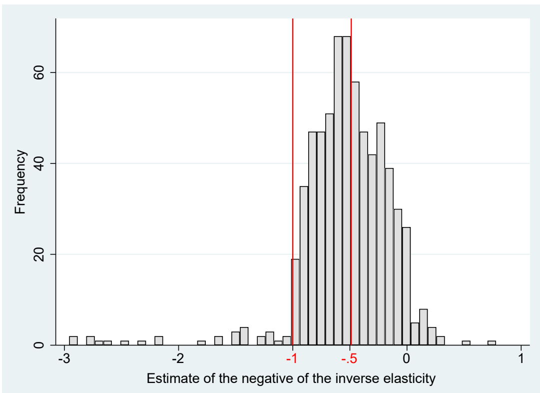
Acemoglu (2002)	D'Amuri *et al.* (2010)	Kesselman *et al.* (1977)
Angrist (1995)	Das (1999)	Kim (2005)
Askilden & Nilsen (2005)	Denny & Fuss (1977)	Kiyota & Kurokawa (2019)
Autor (2014)	Dogan & Akay (2019)	Klenow & Rodriguez-Clare (1997)
Autor *et al.* (2008)	Dougherty (1972)	Klotz *et al.* (1980)
Avalos & Savvides (2006)	Dupuy & Marey (2008)	Krusell *et al.* (2000)
Baum-Snow *et al.* (2018)	Dupuy (2007)	Kwack (2012)
Behar (2010)	Dustmann *et al.* (2009)	Li (2010)
Bergstrom & Panas (1992)	Fallon & Layard (1975)	Lindquist (2005)
Berndt & Christensen (1974)	Fernandez & Messina (2018)	Malmberg (2018)
Berndt & Morrisson (1979)	Fitzgerald & Kearney (2000)	Manacorda *et al.* (2010)
Binelli (2015)	Foldvari & van Leeuwen (2006)	Manacorda *et al.* (2012)
Blankenau & Cassou (2011)	Freeman & Medoff (1982)	McAdam & Willman (2018)
Blundell *et al.* (2016)	Freeman (1975)	Medina & Posso (2010)
Boler (2016)	Gallego (2012)	Mello (2011)
Borghans & ter Weel (2008)	Gancia *et al.* (2013)	Mollick (2008)
Borjas & Katz (2007)	Giannarakis (2017)	Murphy *et al.* (1998)
Borjas (2003)	Glitz & Wissmann (2021)	Nissim (1984)
Borjas *et al.* (2012)	Goldin & Katz (2009)	Ohanian *et al.* (2021)
Bound *et al.* (2004)	Gunadi (2019)	Ottaviano & Peri (2012)
Bowles (1970)	Gyimah-Brempong & Gyapong (1992)	Psacharopoulos & Hinchliffe (1972)
Bowlus *et al.* (2022)	Heckman *et al.* (1998)	Razzak & Timmins (2008)
Brucker & Jahn (2011)	Hendricks & Schoellman (2018)	Reijnders *et al.* (2021)
Busch *et al.* (2020)	Hendricks & Schoellman (2022)	Reshef (2007)
Caliendo *et al.* (2021)	Hijzen *et al.* (2005)	Riano (2009)
Card & Lemieux (2001)	Jamet (2005)	Robbins (1996)
Card (2009)	Jensen & Morrisey (1986)	Silva (2008)
Carneiro *et al.* (2022)	Jerzmanowski & Tamura (2020)	Tinbergen (1974)
Carrasco *et al.* (2015)	Johnson & Keane (2013)	te Velde & Morrissey (2004)
Choi *et al.* (2005)	Johnson (1970)	Verdugo (2014)
Ciccone & Peri (2005)	Katz & Murphy (1992)	Wei *et al.* (2019)
Corker & Bayoumi (1991)	Kawaguchi & Mori (2016)	Welch (1970)
Cruz *et al.* (2020)	Kearney (1997)	Yang (2012)

We search for studies in Google Scholar, which allows our search query to go through the full text of research papers, not just the title, keyword, and abstract, which is the case for most other databases. We examine the first 500 studies returned by the search. We read the abstract of each study to identify those that may potentially include empirical estimates of the elasticity; we then download such studies and read them in detail. Furthermore, we inspect the lists of references of all these studies to find any potentially important papers omitted by our Google Scholar search; we terminate the literature search on October 31, 2021. The data and code are available at `meta-analysis.cz/skill`.

Three co-authors have collected 1/3 of the data each and randomly checked 20% of the data collected by the remaining two co-authors in order to identify and correct potential inconsistencies in coding. The final sample includes 1,096 estimates of the elasticity collected from 99 studies listed in Table A1; we call them primary studies. (Note that only 965 of these estimates are reported together with a standard error, which means that we can use them in publication bias tests.) The oldest study was published in 1970, the most recent is forthcoming as of 2022, covering five decades of research. To give less weight to outliers we winsorize estimates at the 1% level. The histogram of the collected estimates is presented in Figure A1 for both inverse elasticities (from regressions of the skill premium on relative labor supply) and direct elasticities (from reverse regressions and translog specifications, see Appendix B for details). The literature uses both streams of studies for calibration, but since in most situations causality can be expected to run from labor supply changes to changes in the skill premium, interpretation of the directly estimated elasticities is less clear. We collect data from both groups of studies but focus on inverse elasticities in the main analysis. From the histogram we observe that the values of the elasticity that are used for calibration (often between 1 and 2) form a minority of empirical estimates in both groups of studies.

Figure A1: Distribution of the reported estimates





Notes: The figure depicts a histogram of the elasticities reported by individual studies. The vertical lines denote the interval for the elasticity of $\langle 1, 2\rangle$, from which most of the values used for calibrations are drawn. For ease of exposition, outliers are excluded from the figure but included in all statistical tests.

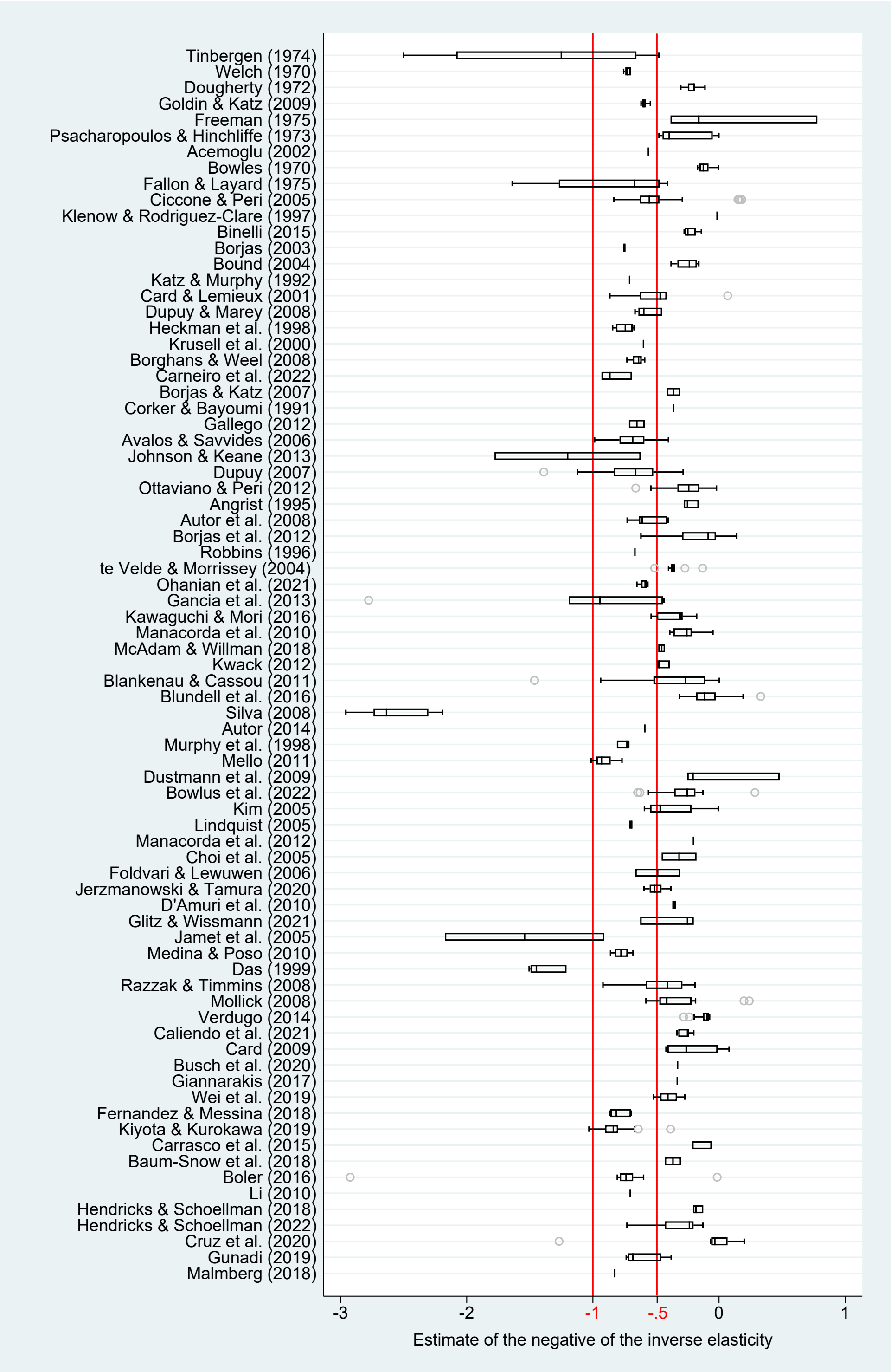
Table A2: Summary statistics for different subsets of the literature

	No. of obs.	Unweighted			Weighted		
		Mean	95% conf. int.		Mean	95% conf. int.	
Direct elasticity estimate	414	0.921	0.788	1.054	1.507	1.343	1.671
Negative inverse elasticity	682	-0.554	-0.589	-0.520	-0.545	-0.582	-0.508
Subsamples of the negative inverse elasticity							
Data characteristics							
Higher frequency	65	-0.263	-0.319	-0.207	-0.277	-0.332	-0.222
Annual frequency	528	-0.576	-0.612	-0.540	-0.564	-0.603	-0.526
Lower frequency	89	-0.639	-0.785	-0.493	-0.893	-1.088	-0.699
Micro data	179	-0.584	-0.655	-0.513	-0.456	-0.519	-0.393
Sectoral data	112	-0.734	-0.863	-0.606	-0.872	-1.027	-0.718
Aggregated data	391	-0.489	-0.523	-0.455	-0.510	-0.548	-0.473
Cross-section	58	-0.635	-0.794	-0.476	-0.631	-0.782	-0.481
Panel or time-series	624	-0.547	-0.582	-0.512	-0.529	-0.566	-0.491
Structural variation							
United States	285	-0.416	-0.452	-0.380	-0.492	-0.532	-0.452
Developed country	443	-0.528	-0.577	-0.479	-0.547	-0.597	-0.497
Developing country	152	-0.474	-0.511	-0.437	-0.476	-0.519	-0.433
Manufacturing sector	95	-0.821	-0.924	-0.717	-0.675	-0.805	-0.545
Regional estimate	93	-0.449	-0.496	-0.402	-0.384	-0.435	-0.332
Country estimate	583	-0.579	-0.618	-0.539	-0.570	-0.612	-0.529
Design of the production function							
One-level CES function	229	-0.465	-0.512	-0.419	-0.479	-0.529	-0.428
Multilevel CES function	453	-0.599	-0.646	-0.553	-0.588	-0.638	-0.537
Estimation technique							
Dynamic model	52	-0.574	-0.717	-0.431	-0.662	-0.828	-0.496
Unit fixed effects	342	-0.669	-0.723	-0.615	-0.637	-0.699	-0.576
Time fixed effects	157	-0.504	-0.564	-0.443	-0.414	-0.472	-0.355
OLS method	362	-0.472	-0.512	-0.432	-0.531	-0.577	-0.484
IV method	264	-0.584	-0.621	-0.547	-0.514	-0.555	-0.473
Natural experiment	40	-1.191	-1.535	-0.847	-0.927	-1.235	-0.618
Publication characteristics							
Unpublished study	202	-0.778	-0.857	-0.699	-0.790	-0.882	-0.698
Published study	480	-0.460	-0.493	-0.427	-0.484	-0.522	-0.447
Top journal publication	131	-0.442	-0.492	-0.393	-0.410	-0.458	-0.362

Notes: The table reports summary statistics of the reported elasticity of substitution for different subsets of the literature and includes also estimates reported without the standard error (which are excluded in the analysis of publication bias). The exact definition of the variables is available in Table D1. Weighted = estimates are weighted by the inverse of the number of estimates reported per study.

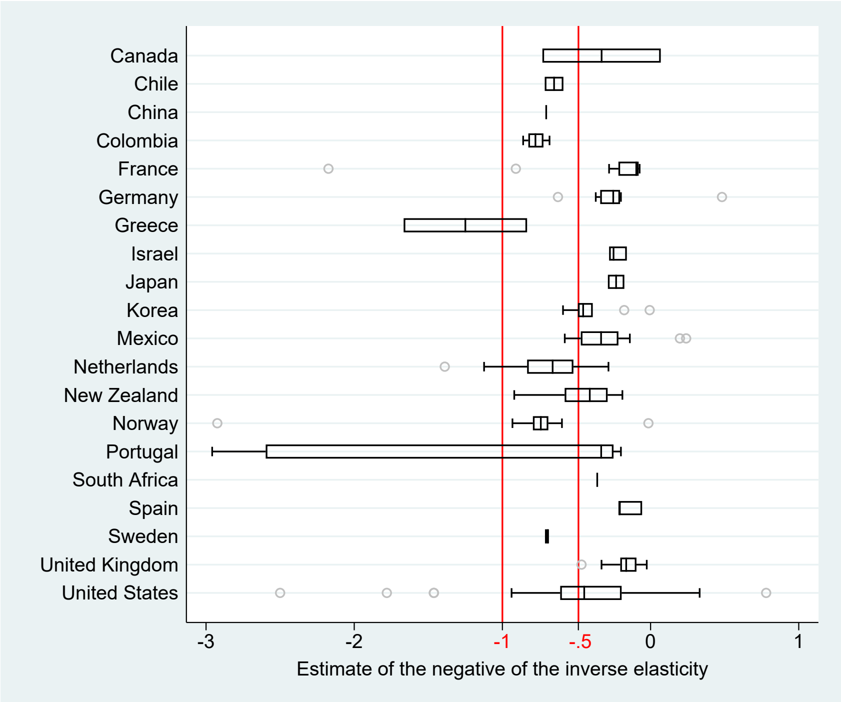
The mean elasticity, directly estimated, is 0.9 (averaged over 414 estimates). The mean estimated negative inverse elasticity is -0.6 (averaged over 682 estimates), which implies an elasticity of 1.8. Summary statistics for various subsamples of the literature are available in Table A2. Figure A2 shows that estimates differ substantially not only across but also within individual studies. Substantial heterogeneity is one stylized fact of the literature on skill substitution. The second stylized fact that arises from a bird's-eye view of the data is the break in the frequency of estimated negative inverse elasticities at -1 and especially at 0 (Figure A1), which might suggest publication bias.

Figure A2: Estimates of the negative inverse elasticity vary both within and across studies



Notes: The studies are sorted by the age of the data they use from oldest to youngest. The length of each box represents the interquartile range (P25-P75), and the dividing line inside the box is the median value. The whiskers represent the highest and lowest data points within 1.5 times the range between the upper and lower quartiles. The vertical lines denote the interval for the elasticity of $\langle 1, 2\rangle$, from which most of the values used for calibrations are drawn. For ease of exposition, outliers are excluded from the figure but included in all statistical tests.

Figure A3: Cross-country heterogeneity in the negative inverse elasticity

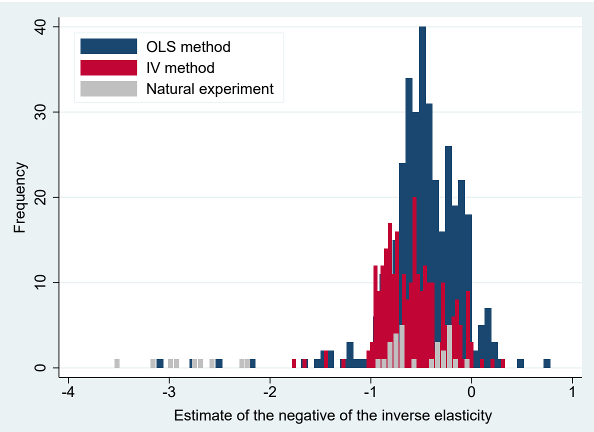


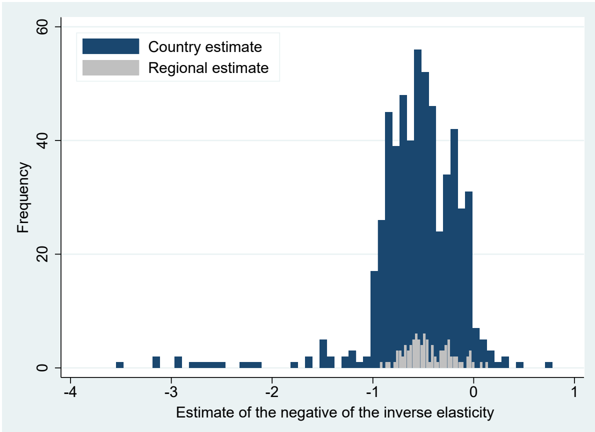
Notes: The length of each box represents the interquartile range (P25-P75), and the dividing line inside the box is the median value. The whiskers represent the highest and lowest data points within 1.5 times the range between the upper and lower quartiles. The vertical lines denote the interval for the elasticity of $\langle 1, 2\rangle$, from which most of the values used for calibrations are drawn. For ease of exposition, outliers are excluded from the figure but included in all statistical tests.

Figure A4: Prima facie patterns in the reported negative inverse elasticities

(a) Estimation approach

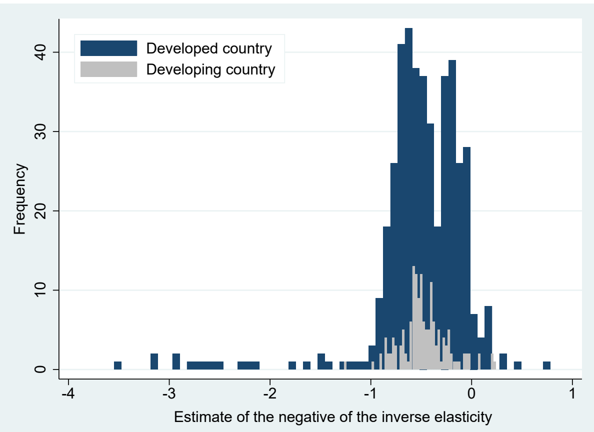
(b) Estimate aggregation

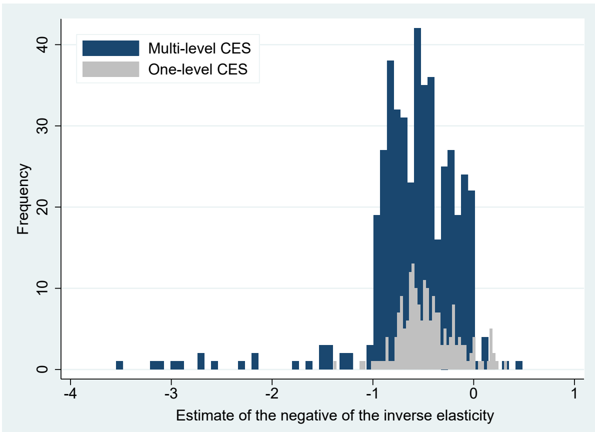




(c) Country classification

(d) Production function





Notes: We use the IMF definition to classify countries as developed or developing.

B Direct Estimates of the Elasticity

As we have noted, most studies identify the elasticity of substitution between skilled and unskilled labor by regressing relative wages on the relative labor supply; the estimated coefficient represents the (negative) inverse elasticity. This approach is intuitive because relative labor supply cannot change fast in reaction to changes in relative wages. If a study is based on genuine random variation in the relative labor supply, then it can identify the causal effect. Other studies estimate the elasticity directly by either running a reverse regression or using a translog specification. These two groups of studies cannot be combined in an analysis of publication bias because the inversion necessary for such a combination creates a mechanical relationship between estimates and standard errors. Therefore the two groups of regression coefficients have to be analyzed separately. In the main body of the paper we focus on the inverse elasticity for three reasons.

First, out of the 99 studies that we find for this meta-analysis, only 24 studies estimate the elasticity directly (including two studies that also report inverse estimates). Second, only 75% of the direct estimates are reported together with standard errors, compared to 96% of the inverse estimates. This observation, especially in the case of studies using the translog function, further decreases the power of publication bias tests for direct estimates. Table B1 shows that studies reporting direct elasticities are also typically older and published in outlets with a smaller impact factor. Third, in most cases we find the identification arguments for reverse regressions less persuasive: it is unclear what can be achieved by regressing the "treatment" (relative labor supply) on the "outcome" (relative wages).

Table B1: Studies relying on direct estimates look worse on paper

Reported standard errors	-0.771**
	(0.377)
Publication year	-0.530**
	(0.224)
Impact factor	-0.393**
	(0.195)
Citations	-0.207
	(0.174)
Constant	2.219**
	(0.887)
Observations (studies)	99

Notes: The table shows the results of a probit regression (response variable = 1 if the study reports only direct estimates of the elasticity). Reported standard errors = 1 if the study reports standard errors, p-values, or t-statistics for any of its point estimates, Publication year in logs, Impact factor is the discounted recursive RePEc impact factor of the outlet, Citations = log of the number of per-year citations of the study in Google Scholar. Robust standard errors in parentheses.

Table B2 lists all the studies that report direct estimates of the elasticity of substitution between skilled and unskilled labor. We divide them into five categories according to the identification arguments that they use: studies using highly disaggregated data, studies introducing a new theory, studies concerned about measurement error, studies using the translog production function, and studies using the translog cost function. Bowles (1970) and Ciccone & Peri (2005) also report, in fact as their key results, estimates of inverse elasticities, so we include those in the main analysis presented in the paper.

Table B2: The 24 studies reporting direct estimates of the elasticity

Category	Study	Justification
Disaggregated	Boler (2016) Freeman & Medoff (1982) Reshef (2007) Riano (2009) Yang (2012)	Estimated for individual firms or small industries, authors assume that the relative price of skilled labor is fixed. Reshef (2007, p. 13): "So it is not a terrible sin to ignore the classic simultaneous demand-supply identification problem for an individual industry."
Theory	Behar (2010) Brucker & Jahn (2011)	The identification approach relies on a concept different from the rest of the literature (monopolistic competition, wage setting, technology import effects). Brucker & Jahn (2011, p. 302): "It follows from our wage-setting framework that labor demand is endogenously determined once wages are fixed."
Measurement error	Bowles (1970) Johnson (1970)	Estimated in response to concerns regarding attenuation bias. Bowles (1970, p. 75): "An upper limit to this bias can be established by making the ratio of labor quantities the dependent variable."
Translog production	Berndt & Christensen (1974) Ciccone & Peri (2005) Denny & Fuss (1977) Jensen & Morrisey (1986)	Relaxing the assumption that the elasticity is constant along the demand curve; authors regress skilled workers' share of wages on relative labor supply. Ciccone & Peri (2005, p. 659): "The key parameter can be estimated consistently using the same instruments and the same identifying assumptions as in the CES case."
Translog cost	Askilden & Nilsen (2005) Bergstrom & Panas (1992) Berndt & Morrisson (1979) Dogan & Akay (2019) Fitzgerald & Kearney (2000) Gyimah-Brempong & Gyapong (1992) Hijzen *et al.* (2005) Kearney (1997) Kesselman *et al.* (1977) Klotz *et al.* (1980) Nissim (1984)	Relaxing the assumption that the elasticity is constant; authors regress skilled workers' share of wages on relative wages (following Shephard's duality theory). Often used with relatively disaggregated data, but we have found no explicit justification of this identification approach in the specific context of skill substitution. In addition, standard errors are rarely reported in this group of studies.

Notes: The table lists studies that report direct estimates of the elasticity of substitution between skilled and unskilled labor—in contrast with the estimates used in our main analysis, which need to be inverted to yield the elasticity (the standard approach in the literature). Because inversion creates a mechanical relationship between estimates and standard errors, these two groups of studies cannot be pooled together in an analysis of publication bias. Note that Bowles (1970) and Ciccone & Peri (2005) also report, in fact as their main results, inverse elasticities, so we include those in the main analysis.

Regarding disaggregated data, five studies make, explicitly or implicitly, the assumption that their dataset is disaggregated enough to allow them analyze firm-level demand for skills. Under perfect competition, the relative price of skill is assumed to be fixed. A persuasive dataset and identification strategy are provided by Boler (2016) for the 2002 R&D expenditures reform in Norway, but for the remaining studies (three of them unpublished, the remaining one published in 1982) identification is less clear.

Regarding studies presenting a new theory, Brucker & Jahn (2011) assume monopolistic competition and wage-setting framework, in which wages are fixed first and labor outcomes are determined later. The unpublished paper by Behar (2010) features a standard regression of the wage premium on the relative relative labor supply but the author interprets the regression coefficient as a direct estimate of the elasticity based on a theoretical framework that incorporates technology import effects for developing countries.

Regarding measurement error, Bowles (1970) estimates a reverse regression to establish an upper bound for the extent of attenuation bias but otherwise focuses on a standard regression of the wage premium on the relative labor supply. The motivation for the use of reverse regression by Johnson (1970) is less clear but implicitly seems to be based on the idea that the wage premium is measured with less noise.

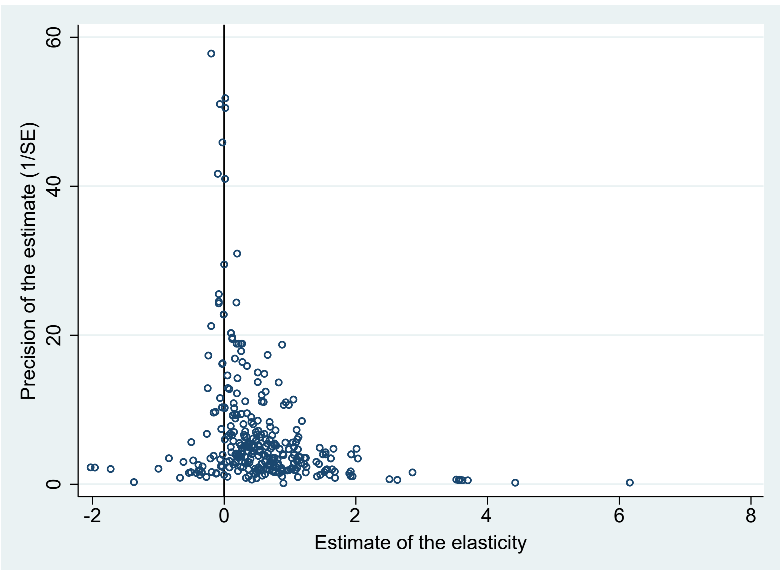
Regarding the translog production function, its use is motivated by the desire to relax the assumption that the elasticity is constant along the demand curve. Researchers regress skilled workers' share of wages on the relative labor supply; a more detailed description is available in the informative paper by Ciccone & Peri (2005, pp. 659–661), who also estimate the inverse elasticity. Hence the identification argument for the translog production function is persuasive, at least in the case of Ciccone & Peri (2005), and generally similar to the standard case for inverse estimates—but here the regression coefficient does not need to be inverted to yield the elasticity and thus cannot be pooled with the dataset used in the main body of our paper. The subgroup is unfortunately too small to be analyzed separately.

Regarding the translog cost function, this is the most common approach for the direct estimation of the elasticity of substitution: 11 studies in our dataset use this technique. The motivation is similar to the one for the translog production function, but here researchers regress skilled workers' share of wages on relative wages, so identification resembles reverse regressions. We failed to find any explicit justification of this identification approach in the specific context of skill substitution. The studies in this group often use relatively disaggregated data, so they might implicitly rely on the same assumption as the first group of studies described above. In addition, estimates from both translog cost and production functions (given the flexibility of the translog function, often means or medians of many potential values) are less commonly accompanied by standard errors or other statistics from which standard errors can be computed, which effectively prevents us from attempting any analysis of publication bias for this subgroup.

C Additional Material: Publication Bias

Here we present results for the sample of direct estimates, additional bias-correction techniques (for context, they are reported and discussed together with the techniques introduced in the main text), results for various subsets of the inverse estimates, and tests related to the assumption of conditional independence of estimates and standard errors. The additional techniques that we use are OLS meta-regression, the weighted average of adequately powered estimates (WAAP) by Ioannidis *et al.* (2017), and the stem-based technique by Furukawa (2020). WAAP first roughly estimates the underlying effect, then computes retrospective power for each reported estimate, drops estimates with less than 80% power, and computes the average of the remaining estimates weighted by inverse variance. The question is what rough estimate to choose in the first stage, and we follow the baseline specification of Ioannidis *et al.* (2017) by selecting the mean weighted by inverse variance. The stem-based technique builds on the funnel plot and exploits the trade-off between variance and bias: when only the most precise studies are included to compute the mean, publication bias is small, but it is inefficient to discard so many studies. Furukawa (2020) shows how to optimally balance bias and variance.

Figure C1: The funnel plot suggests publication bias among direct estimates



Notes: When there is no publication bias, the funnel plots should be symmetrical. Outliers are excluded from the figure for ease of exposition but included in all statistical tests.

Figure C1 shows evidence of asymmetry in the funnel plot for direct estimates, a finding consistent with publication bias against negative and insignificant estimates of the elasticity. Table C1 shows the results of publication bias tests for direct and inverse elasticities. Regarding direct elasticities, all techniques find evidence of publication bias or, if they do not provide tests of the bias (WAAP and Stem), find corrected mean estimates much smaller than the uncorrected mean of 0.9. The use of study-level fixed or between effects does not change the conclusion. In the IV specification the instrument for the standard error is weak (the first-stage robust F-statistic is 6), so we report the two-step weak-instrument 95% confidence interval based on Andrews (2018). The overall message, consistent with the funnel plot, is that the corrected direct elasticity is zero. The only exception is the selection model by Andrews & Kasy (2019), which is more conservative in publication bias correction and suggests a mean of 0.4. But even this model finds that estimates significant at the 5% level are about twice as likely to be reported than insignificant estimates, and the implied exaggeration due to publication bias is more than twofold.

Regarding inverse elasticities, our results are similar. Publication bias is strong, and the corrected inverse elasticity is close to zero with the exception of the Andrews & Kasy (2019) model: even here, however, the implied elasticity of substitution exceeds 3. Note that the results for direct and inverse estimates are mutually inconsistent because they imply, respectively, zero and infinite elasticity of substitution. The identification assumptions in one of the streams of the literature are thus likely violated; as we explain in Appendix B, we believe this is the case for most of the direct estimates. Another potential explanation is that both streams of the literature are identified reasonably well but attenuation bias drives the estimated coefficients to zero. While the sample of direct estimates is too limited to allow a meaningful analysis of attenuation bias, we find evidence consistent with the bias in the sample of inverse elasticities.

For the remaining analysis we only consider inverse estimates. Table C2 shows the results for the subsamples of OLS, IV, and natural experiments. The addition of OLS, WAAP and Stem techniques does not change the conclusions described in the main text (only the Stem method never finds a statistically significant corrected inverse elasticity). Unfortunately we only have 40 estimates from 6 natural experiments, so the power of the tests is low, but all techniques suggest strong publication bias and negligible corrected effects. Natural experiments are thus consistent with no causal effect of relative skill supply on the skill premium and therefore with infinite elasticity of substitution. We obtain similar results for OLS estimates—with the exception of the Andrews & Kasy (2019) model, which is once again less aggressive in correcting for publication bias. IV estimates of the negative inverse elasticity are different: they show less publication bias and more negative inverse elasticities, implying an elasticity around 4. The results are consistent with attenuation bias in the literature (IV estimates of inverse elasticities are larger in magnitude than OLS estimates) and little additional endogeneity bias (OLS estimates are similar to estimates from natural experiments).

In Table C3, Table C4, and Table C5 we examine other subsamples: developed countries vs. developing countries, elasticities estimated at the country level vs. at the regional level, one-level CES functions vs. a multilevel CES functions. Once again the addition of

OLS, WAAP and Stem does not change our conclusions. The results suggest that elasticities tend to be larger for developed countries (above 4) than developing countries (around 2.5), and publication bias is stronger for the former group, which displays a corrected inverse elasticity closer to zero. Next, our results suggest that elasticities estimated at the country level are smaller than those estimated at the regional level, but we only have 93 estimates for the latter group. Finally, both one-level and multilevel CES functions seem to yield similar elasticities.

Table C1: Tests point to strong publication bias, small corrected coefficients

Part 1: Direct elasticity				
Panel A: linear	OLS	FE	BE	IV
Publication bias	2.323***	1.574*	2.929***	1.534***
(*Standard error*)	(0.422)	(0.833)	(0.761)	(0.427)
	[-0.812, 5.227]			[0.919, 3.591]
				{0.459, 7.568}
Effect beyond bias	-0.0191	0.0906	0.0197	0.0963
(*Constant*)	(0.0843)	(0.122)	(0.0735)	(0.113)
	[-1.289, 0.524]			[-0.655, 0.341]
First-stage robust F-stat				6.05
Observations	311	311	311	309
Panel B: nonlinear	WAAP	Stem method	Endog. kink	Selection model
Publication bias			2.333***	P=0.561
			(0.212)	(0.186)
Effect beyond bias	-0.0414	-0.0581	-0.0242	0.403**
	(0.0312)	(0.144)	(0.0195)	(0.205)
Observations	311	311	311	311
Part 2: Negative inverse elasticity				
Panel A: linear	OLS	FE	BE	IV
Publication bias	-3.751***	-4.530***	-2.736***	-4.491***
(*Standard error*)	(0.735)	(1.311)	(0.781)	(1.161)
	[-5.54, -2.106]			[-6.985, -1.984]
				{-9.229, -2.127}
Effect beyond bias	-0.0822	-0.0259	-0.162***	-0.0491
(*Constant*)	(0.0781)	(0.0948)	(0.0470)	(0.103)
	[-0.348, 0.101]			[-0.381, 0.141]
First-stage robust F-stat				14.28
Observations	654	654	654	505
Panel B: nonlinear	WAAP	Stem method	Endog. kink	Selection model
Publication bias			-3.796***	P=0.385
			(0.296)	(0.110)
Effect beyond bias	-0.161***	0.102	-0.0768***	-0.300***
	(0.0220)	(0.208)	(0.0135)	(0.083)
Observations	654	654	654	654

Notes: Specifications in Panel A regress estimates on standard errors (weighted by inverse variance). Standard errors, clustered at the study level, are in parentheses. 95% confidence intervals from wild bootstrap (Roodman *et al.*, 2018) are in square brackets. FE = study fixed effects. BE = study between effects. IV = the inverse of the square root of the number of observations is used as an instrument for the standard error. In curly brackets we show the two-step weak-instrument-robust 95% confidence interval based on Andrews (2018) and Sun (2018). WAAP = weighted average of adequately powered estimates (Ioannidis *et al.*, 2017), Stem = the method by Furukawa (2020), Endog. kink = the method by Bom & Rachinger (2019), Selection model = the method by Andrews & Kasy (2019), P denotes the probability that estimates insignificant at the 5% level are published relative to the probability that significant estimates are published (normalized at 1). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C2: IV estimation of the inverse elasticity shows less bias and a larger corrected effect

Part 1: OLS estimates of the inverse elasticity				
Panel A: linear	OLS	FE	BE	IV
Publication bias	-5.393***	-5.804***	-4.277***	-6.962***
(*Standard error*)	(0.856)	(1.999)	(1.266)	(1.694)
	[-7.072, -3.165]			[-11.770, -2.494]
				{-11.972, -3.133}
Effect beyond bias	-0.0420	-0.0207	-0.0965	0.0103
(*Constant*)	(0.0757)	(0.103)	(0.0627)	(0.104)
	[-0.339, 0.134]			[-0.331, 0.214]
First-stage robust F-stat				46.17
Observations	347	347	347	251
Panel B: nonlinear	WAAP	Stem method	Endog. kink	Selection model
Publication bias			-5.465***	P=0.468
			(0.540)	(0.139)
Effect beyond bias	-0.144**	0.102	-0.0361**	-0.289**
	(0.0252)	(0.201)	(0.0191)	(0.113)
Observations	347	347	347	347
Part 2: IV estimates of the inverse elasticity				
Panel A: linear	OLS	FE	BE	IV
Publication bias	-1.489***	-2.287**	-0.923	-0.553
(*Standard error*)	(0.577)	(0.843)	(1.365)	(0.681)
	[-2.962, 0.913]			[-1.913, 1.078]
				{-1.991, 0.748}
Effect beyond bias	-0.252**	-0.149	-0.297**	-0.400***
(*Constant*)	(0.123)	(0.109)	(0.115)	(0.114)
	[-0.561, 0.046]			[-0.719, 0.175]
First-stage robust F-stat				69.98
Observations	264	264	264	212
Panel B: nonlinear	WAAP	Stem method	Endog. kink	Selection model
Publication bias			-1.485***	P=0.336
			(0.268)	(0.093)
Effect beyond bias	-0.330***	-0.151	-0.252***	-0.333***
	(0.0308)	(0.195)	(0.0246)	(0.058)
Observations	264	264	264	264

Continued on next page

Table C2: IV estimation of the inverse elasticity shows less bias and a larger corrected effect (continued)

Part 3: Natural experiments				
Panel A: linear	OLS	FE	BE	IV
Publication bias	-3.282***	-3.557***	-1.874*	-3.176***
(*Standard error*)	(0.880)	(0.0178)	(0.682)	(0.853)
	[-4.801, -1.610]			[-4.854, -1.407]
				{-4.653, -1.444}
Effect beyond bias	0.0116	0.0496***	-0.121	-0.00307
(*Constant*)	(0.0322)	(0.00246)	(0.0824)	(0.0297)
	[-0.672, 0.810]			[NA, NA]
First-stage robust F-stat				260.41
Observations	40	40	40	40
Panel B: nonlinear	WAAP	Stem method	Endog. kink	Selection model
Publication bias			-3.115***	P=0.187
			(0.343)	(0.075)
Effect beyond bias	-0.0146	-0.162	0.00302	-0.009
	(NA)	(0.162)	(0.0280)	(0.066)
Observations	40	40	40	40

Notes: Specifications in Panel A regress estimates on standard errors (weighted by inverse variance). Standard errors, clustered at the study level, are in parentheses. 95% confidence intervals from wild bootstrap (Roodman *et al.*, 2018) are in square brackets. FE = study fixed effects. BE = study between effects. IV = the inverse of the square root of the number of observations is used as an instrument for the standard error. In curly brackets we show the two-step weak-instrument-robust 95% confidence interval based on Andrews (2018) and Sun (2018). WAAP = weighted average of adequately powered estimates (Ioannidis *et al.*, 2017), Stem = the method by Furukawa (2020), Endog. kink = the method by Bom & Rachinger (2019), Selection model = the method by Andrews & Kasy (2019), P denotes the probability that estimates insignificant at the 5% level are published relative to the probability that significant estimates are published (normalized at 1). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C3: Publication bias tests for subsamples of inverse elasticities estimated for developed and developing countries

Part 1: Inverse elasticities estimated for developed countries				
Panel A: linear	OLS	FE	BE	IV
Publication bias	-3.744***	-4.526***	-2.748***	-5.391***
(*Standard error*)	(0.703)	(1.320)	(0.634)	(1.357)
	[-5.371, -2.073]			[-8.829, -2.605]
				{-14.237, -2.345}
Effect beyond bias	-0.0323	0.0194	-0.122***	0.0674
(*Constant*)	(0.0744)	(0.0875)	(0.0372)	(0.0675)
	[-0.323, 0.151]			[-0.244, 0.203]
First-stage robust F-stat				8.72
Observations	418	418	418	299
Panel B: nonlinear	WAAP	Stem method	Endog. kink	Selection model
Publication bias			-3.771***	P=0.433
			(0.349)	(0.109)
Effect beyond bias	-0.0838***	0.102	-0.0276*	-0.258***
	(0.0285)	(0.171)	(0.0141)	(0.088)
Observations	418	418	418	418
Part 2: Inverse elasticities estimated for developing countries				
Panel A: linear	OLS	FE	BE	IV
Publication bias	-0.456	-1.204	0.581	0.193
(*Standard error*)	(0.690)	(0.803)	(1.913)	(1.023)
	[-1.679, 2.875]			[-2.334, 5.368]
				{-1.743, 2.102}
Effect beyond bias	-0.404***	-0.349***	-0.478***	-0.457***
(*Constant*)	(0.0803)	(0.0593)	(0.149)	(0.0546)
	[-0.804, -0.0323]			[-0.837, -0.397]
First-stage robust F-stat				275.96
Observations	151	151	151	128
Panel B: nonlinear	WAAP	Stem method	Endog. kink	Selection model
Publication bias			-0.453	P=0.654
			(0.446)	(0.442)
Effect beyond bias	-0.423***	-0.391***	-0.404***	-0.425***
	(0.0188)	(0.0731)	(0.0261)	(0.030)
Observations	151	151	151	151

Notes: Specifications in Panel A regress estimates on standard errors (weighted by inverse variance). Standard errors, clustered at the study level, are in parentheses. 95% confidence intervals from wild bootstrap (Roodman *et al.*, 2018) are in square brackets. FE = study fixed effects. BE = study between effects. IV = the inverse of the square root of the number of observations is used as an instrument for the standard error. In curly brackets we show the two-step weak-instrument-robust 95% confidence interval based on Andrews (2018) and Sun (2018). WAAP = weighted average of adequately powered estimates (Ioannidis *et al.*, 2017), Stem = the method by Furukawa (2020), Endog. kink = the method by Bom & Rachinger (2019), Selection model = the method by Andrews & Kasy (2019), P denotes the probability that estimates insignificant at the 5% level are published relative to the probability that significant estimates are published (normalized at 1). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C4: Publication bias tests for subsamples of inverse elasticities estimated at the country and regional level

Part 1: Inverse elasticities estimated at the country level				
Panel A: linear	OLS	FE	BE	IV
Publication bias	-2.861***	-2.950***	-2.830***	-2.870**
(*Standard error*)	(0.689)	(0.907)	(0.878)	(1.207)
	[-4.799, -1.05]			[-5.694, 0.229]
				{-3.971, -1.146}
Effect beyond bias	-0.193**	-0.187***	-0.188***	-0.238***
(*Constant*)	(0.0753)	(0.0674)	(0.0531)	(0.0916)
	[-0.437, -0.045]			[-0.448, -0.063]
First-stage robust F-stat				7.87
Observations	555	555	555	406
Panel B: nonlinear	WAAP	Stem method	Endog. kink	Selection model
Publication bias			-2.828***	P=0.436
			(0.302)	(0.101)
Effect beyond bias	-0.281***	-0.0884	-0.194***	-0.311***
	(0.0175)	(0.0923)	(0.0151)	(0.096)
Observations	555	555	555	555
Part 2: Inverse elasticities estimated at the regional level				
Panel A: linear	OLS	FE	BE	IV
Publication bias	-2.442***	-2.153***	-2.098***	-2.482***
(*Standard error*)	(0.304)	(0.480)	(0.400)	(0.280)
	[-3.989, -0.991]			[-3.640, -1.121]
				{-3.213, -1.662}
Effect beyond bias	-0.0301	-0.0563	-0.0299	-0.0265***
(*Constant*)	(0.0186)	(0.0434)	(0.0193)	(0.00791)
	[-0.121, 0.155]			[-0.055, -0.004]
First-stage robust F-stat				52.11
Observations	93	93	93	93
Panel B: nonlinear	WAAP	Stem method	Endog. kink	Selection model
Publication bias			-2.440***	P=0.105
			(0.198)	(0.059)
Effect beyond bias	-0.0618**	-0.0668	-0.0304***	-0.162
	(0.0233)	(0.0793)	(0.0102)	(0.114)
Observations	93	93	93	93

Notes: Specifications in Panel A regress estimates on standard errors (weighted by inverse variance). Standard errors, clustered at the study level, are in parentheses. 95% confidence intervals from wild bootstrap (Roodman *et al.*, 2018) are in square brackets. FE = study fixed effects. BE = study between effects. IV = the inverse of the square root of the number of observations is used as an instrument for the standard error. In curly brackets we show the two-step weak-instrument-robust 95% confidence interval based on Andrews (2018) and Sun (2018). WAAP = weighted average of adequately powered estimates (Ioannidis *et al.*, 2017), Stem = the method by Furukawa (2020), Endog. kink = the method by Bom & Rachinger (2019), Selection model = the method by Andrews & Kasy (2019), P denotes the probability that estimates insignificant at the 5% level are published relative to the probability that significant estimates are published (normalized at 1). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C5: Publication bias tests for subsamples of inverse elasticities estimated using one-level and multilevel CES functions

Part 1: Inverse elasticities estimated using one-level CES funtions				
Panel A: linear	OLS	FE	BE	IV
Publication bias	-4.815***	-5.376***	-2.066	-6.121***
(*Standard error*)	(0.806)	(1.156)	(1.318)	(1.697)
	[-6.282, -1.797]			[-14.650, -2.333]
				{-9.133, -2.670}
Effect beyond bias	0.0695	0.114	-0.199*	0.197***
(*Constant*)	(0.126)	(0.0923)	(0.103)	(0.0234)
	[-0.739, 2.054]			[-3.071, 1.167]
First-stage robust F-stat				6,322.62
Observations	198	198	198	149
Panel B: nonlinear	WAAP	Stem method	Endog. kink	Selection model
Publication bias			-4.903***	P=0.507
			(0.549)	(0.191)
Effect beyond bias	0.144***	0.130	0.0778***	-0.371***
	(0.0209)	(0.152)	(0.0256)	(0.118)
Observations	198	198	198	198
Part 2: Inverse elasticities estimated using multilevel CES funtions				
Panel A: linear	OLS	FE	BE	IV
Publication bias	-3.289***	-3.370***	-3.331***	-3.345***
(*Standard error*)	(0.685)	(0.744)	(1.086)	(1.031)
	[-5.355, -1.709]			[-5.774, -.3949]
				{-4.595, -1.942}
Effect beyond bias	-0.144**	-0.139***	-0.145**	-0.177**
(*Constant*)	(0.0591)	(0.0513)	(0.0559)	(0.0863)
	[-0.372, -0.034]			[-0.465, -0.021]
First-stage robust F-stat				11.49
Observations	444	444	444	348
Panel B: nonlinear	WAAP	Stem method	Endog. kink	Selection model
Publication bias			-3.274***	P=0.350
			(0.328)	(0.090)
Effect beyond bias	-0.161***	-0.0860	-0.144***	-0.261**
	(0.0220)	(0.174)	(0.0147)	(0.104)
Observations	444	444	444	444

Notes: Specifications in Panel A regress estimates on standard errors (weighted by inverse variance). Standard errors, clustered at the study level, are in parentheses. 95% confidence intervals from wild bootstrap (Roodman *et al.*, 2018) are in square brackets. FE = study fixed effects. BE = study between effects. IV = the inverse of the square root of the number of observations is used as an instrument for the standard error. In curly brackets we show the two-step weak-instrument-robust 95% confidence interval based on Andrews (2018) and Sun (2018). WAAP = weighted average of adequately powered estimates (Ioannidis *et al.*, 2017), Stem = the method by Furukawa (2020), Endog. kink = the method by Bom & Rachinger (2019), Selection model = the method by Andrews & Kasy (2019), P denotes the probability that estimates insignificant at the 5% level are published relative to the probability that significant estimates are published (normalized at 1). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C6: Regressing estimates on standard errors when p-value < 0.005

	All inverse	OLS method	IV method	Natural experiment	Developed country
Standard error	-4.864***	-6.208***	-0.947	-5.379***	-4.903***
	(0.941)	(1.227)	(0.922)	(0.368)	(0.816)
	[-7.376, -2.372]	[-9.257, -3.210]	[-2.848, 3.234]	[-6.323, -4.835]	[-6.902, -3.053]
Observations	368	237	115	15	222
	Developing country	Country estimate	Region estimate	One-level CES function	Multilevel CES function
Standard error	-0.848	-4.907***	-3.433***	-1.534	-5.339***
	(1.013)	(1.143)	(0.188)	(1.802)	(1.025)
	[-2.295, 4.223]	[-8.007, -2.069]	[-4.908, -1.406]	[-8.816, 2.251]	[-8.110, -2.947]
Observations	94	333	35	101	264

Notes: The response variable is the estimate of the negative inverse elasticity. The constant is included in the regressions but not reported in the table. Standard errors, clustered at the study level, are shown in parentheses. 95% confidence intervals from wild bootstrap (Roodman *et al.*, 2018) are in square brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C7: Specification test for the Andrews & Kasy (2019) model

	All inverse	OLS method	IV method	Natural experiment	Developed country
Correlation	0.515	0.265	0.635	0.947	0.558
	[0.436, 0.605]	[0.160, 0.392]	[0.499, 0.743]	[0.928, 0.976]	[0.462, 0.667]
Observations	654	347	264	40	418
	Developing country	Country estimate	Region estimate	One-level CES function	Multilevel CES function
Correlation	0.330	0.856	0.476	0.199	0.592
	[0.159, 0.482]	[0.775, 0.95]	[0.384, 0.568]	[0.047, 0.342]	[0.498, 0.709]
Observations	151	555	93	198	444

Notes: Following Kranz & Putz (2022), the table shows, for various subsets of the literature, the correlation coefficient between the logarithm of the absolute value of the estimated inverse elasticity and the logarithm of the corresponding standard error, weighted by the inverse publication probability estimated by the Andrews & Kasy (2019) model. If the assumptions of the model hold, the correlation is zero. Bootstrapped 95% confidence interval in parentheses.

D Additional Material: Heterogeneity

The studies estimating the elasticity differ in so many dimensions that it is unfeasible to control for all potential differences. In the publication bias section we used study-level fixed effects, which capture study idiosyncrasies but not the characteristics of individual estimation specifications. At the risk that we still omit some characteristics others would find relevant—the list of potential ones is unlimited—, we identify 28 main characteristics (and consequently, to avoid the dummy trap, codify 24 explanatory variables to be used in model averaging) which we distribute for ease of exposition into five categories: data characteristics, structural variation, design of the production function, estimation technique, and publication characteristics. Table D1 lists all the codified characteristics, provides their definitions, and gives summary statistics including the simple mean, standard deviation, and mean weighted by the inverse of the number of estimates reported in a study. Given the number of estimates that we collect, the construction of the dataset required manual collection of about 30,000 data points by three of the co-authors upon carefully reading the primary studies.

Table D1: Description and summary statistics of regression variables

Variable	Description	Mean	SD	WM
Inverse elasticity	Estimate of the negative of the inverse elasticity of substitution between the skilled and unskilled labor (response variable).	-0.543	0.436	-0.515
Standard error (SE)	Standard error of the estimated inverse elasticity. The variable is important for gauging publication bias.	0.190	0.240	0.184
Data characteristics				
Higher frequency	=1 if higher than annual frequency of the data is used; typically monthly, quarterly, or semi-annual.	0.083	0.275	0.134
Annual frequency	=1 if annual frequency of the data is used in the estimation (reference category for data frequency).	0.784	0.412	0.796
Lower frequency	=1 if lower than annual frequency of the data is used; typically three, five, or ten years.	0.133	0.340	0.070
Micro data	=1 if micro-level data (unit = single worker or firm) are used in the estimation.	0.257	0.437	0.267
Sectoral data	=1 if sector-level data (unit = sector) are used in the estimation.	0.164	0.370	0.122
Aggregated data	=1 if aggregated data (unit = economy or regions) are used in the estimation (reference category for the type of data aggregation).	0.580	0.494	0.611
Cross-section	=1 if cross-sectional data are used; =0 if time-series or panel data are used.	0.076	0.266	0.138
Structural variation				
United States	=1 if the country for which the elasticity is estimated is the United States.	0.410	0.492	0.422

Continued on next page

Table D1: Description and summary statistics of regression variables (continued)

Variable	Description	Mean	SD	WM
Developing country	=1 if a developing country is considered.	0.231	0.422	0.177
Manufacturing sector	=1 if the elasticity is estimated for the manufacturing sector, =0 if another sector is considered.	0.145	0.353	0.058
Design of production function				
One-level CES function	=1 if a one-level CES form of the production function is used in the estimation (reference category for the functional form).	0.321	0.467	0.378
Multilevel CES function	=1 if a multilevel CES form of the production function is used in the estimation.	0.679	0.467	0.622
Time control	=1 if time control is included in the model (capturing, e.g., technological change).	0.544	0.498	0.640
Location control	=1 if location/unit control is included (capturing spatial variation).	0.142	0.350	0.149
Macro control	=1 if macroeconomic indicators are included.	0.086	0.280	0.084
Age control	=1 if a control for the age of workers is included.	0.098	0.297	0.179
Capital control	=1 if a capital-related control is included (capturing changes in capital stock under a capital-skill complementarity technology).	0.214	0.410	0.148
Estimation technique				
Dynamic model	=1 if the model form used for estimation is dynamic (VAR, ECM, VECM, PAD, ADL, DLTM, DOLS).	0.076	0.266	0.168
Unit fixed effects	=1 if the model is estimated in first differences or cross-sectional fixed effects are considered.	0.517	0.500	0.430
Time fixed effects	=1 if time fixed effects are included in the model.	0.240	0.427	0.165
OLS method	=1 if the ordinary least squares method or its variations (LS, DOLS, WLS, GLS) are used for estimation (reference category for the method variables).	0.531	0.499	0.611
IV method	=1 if instrumental variables are used, including 2SLS, 3SLS, and GMM.	0.404	0.491	0.270
Natural experiment	=1 if the study uses data from a natural experiment (e.g., has arguably exogenous variation in the relative supply of skilled labor).	0.061	0.240	0.089
Publication characteristics				
Impact factor	The discounted recursive RePEc impact factor of the outlet.	0.846	1.169	1.385
Citations	The logarithm of the number of per-year citations of the study in Google Scholar.	1.612	1.434	2.126

Notes: Table only includes estimates of the inverse elasticity for which standard errors are reported. SD = standard deviation, WM = mean weighted by the inverse of the number of estimates reported per study, CES = constant elasticity of substitution, VAR = vector autoregression, ECM = error correction model, VECM = vector error correction model, PAD = partial adjustment model, ADL = autoregressive distributed lag model, DLTM = distributed lag and trend model, DOLS = dynamic ordinary least squares, WLS = weighted least squares, GLS = generalized least squares, 2SLS = two-stage least squares, 3SLS = three-stage least squares, GMM = generalized method of moments.

Data characteristics The studies in our sample differ in the type of data used to produce estimates of the elasticity. An important aspect is data frequency. With higher frequencies, transitory variation is often present and, if not accounted for, it can generate a biased estimate of the long-run elasticity of substitution (Chirinko & Mallick, 2017). Four fifths of the estimates in our sample employ *annual* data; higher frequencies such as monthly, quarterly, or semi-annual appear relatively scarcely. Another challenge that the researchers have to face is that of data aggregation. Hamermesh (1996) classifies empirical studies into three main groups based on the level of data aggregation. First, there are studies using *aggregated data*, where the unit of observation is the economy or region. Second, aggregation can be conducted at the level of industries (captured by the variable *Sectoral data*). The third group consists of studies where firms or individuals are used as units of observation (*Micro data*).

There are several potential problems with data aggregation. For instance, Hamermesh (1996) criticizes the use of linear aggregation techniques for aggregating nonlinear relationships. Even with the assumption of identical technologies in all firms, one cannot be sure that parameters in the estimated equations are the same for the particular firm and for the aggregated case. Moreover, aggregating workers into groups means implicitly assuming that these workers are very close p-substitutes or q-complements. Furthermore, Broadstock *et al.* (2007) warn that the estimated elasticity involving an aggregate is not necessarily a weighted average of the elasticities for the disaggregated inputs. A practical issue with aggregated data is that fewer observations used in regressions are usually linked with lower precision and that measurement error can differ from that in disaggregated data, which has consequences for both publication and attenuation bias. Another important aspect of data is their dimension: if purely *cross-sectional data* are used or if the time dimension is also taken into account. Hamermesh (1996, p. 63) notes that "there is nothing inherently more attractive in cross-sections or time-series data. Rather, the choice depends on the degree of spatial aggregation in each type of available data." In practice, time series at the micro level are rare, and cross-sectional data generally enable greater disaggregation.

Structural variation Inherent differences in the elasticity among countries and sectors could give rise to another source of heterogeneity. A large part of our sample consists of elasticities computed for the United States (about 40%). The strong consensus in the literature about the elasticity lying between 1 and 2 is to a large extent derived from the US studies by Katz & Murphy (1992), Ciccone & Peri (2005), Autor *et al.* (2008), and Goldin & Katz (2009). Evidence on structural variation has been rather rare in the literature. Psacharopoulos & Hinchliffe (1972) report larger estimates for developed countries compared to developing ones, while Tinbergen (1974) finds the values between 0.4 and 2 for both developing and developed countries. Later studies on *Developing countries*, such as Behar (2010) or Manacorda *et al.* (2010), suggest values between 2 and 4. On balance, according to our reading of the literature the prevailing view is still that the elasticity is larger in more developed countries (Foldvari & van Leeuwen, 2006).

Some authors, such as Blankenau & Cassou (2011), suggest that manufacturing and skilled services (financial or health services, for example) often stand out in the industry-specific analyses. These sectors, with a shifting demand to skilled labor and heavier on specific skill-sets (Berman *et al.*, 1994), may display structurally different elasticities. We create a separate dummy for manufacturing, for which we have enough observations.

Design of the production function Researchers typically assume *one-level CES* (constant elasticity of substitution) production function. But other functional forms are used as well, including most prominently the multilevel (or nested) CES function. For the sake of simplicity, some authors consider solely equation (2), which treats skilled and unskilled labor as the only factors of production. In this form the elasticity is constant irrespective of changes in relative labor supply (Ciccone & Peri, 2005) and can be derived from the parameter ρ as $\sigma = 1/(1-\rho)$. Under the CES framework, more production factors can be nested (*Multilevel CES*) and there are many ways to do so.

Most often, three production factors are considered in estimation: skilled labor, unskilled labor, and capital. One stream of the literature assumes production to be a CES function of capital and labor at the first level and further decomposes labor into skilled and unskilled at the second level via the Cobb-Douglas specification (therefore, the elasticity of substitution between capital and labor is restricted to one; Avalos & Savvides, 2006). Another stream of the literature assumes a CES function with capital and labor at the first level and further decomposes labor into skilled and unskilled parts at the second level via another CES specification (as in Borjas, 2003; Borjas & Katz, 2007). Finally, some studies apply alternative nesting schemes with a CES function of capital, skilled labor, and unskilled labor at the first level; at the second level, skilled workers are divided into more groups according to their specific skills via a CES specification (Manacorda *et al.*, 2010). Among the more complex nesting structures is the one used by Krusell *et al.* (2000) and followed by Lindquist (2005) and Dupuy (2007). They employ four production factors (capital structure, capital equipment, skilled labor, and unskilled labor) and a three-level nesting structure.

Multiple control variables are commonly employed in the basic specification of the production functions described above. These variables capture different characteristics of either workers or labor markets. The most frequent one is *time control* capturing potential technological changes that affect the demand for skills; these controls are used in about half of the regressions in our sample. Other variables control for the location (Acemoglu, 2002), different macroeconomic circumstances such as the level of minimum wage, unemployment rate, and labor market reforms (Manacorda *et al.*, 2010; Autor *et al.*, 2008), and socioeconomic factors such as city size, college share, and union membership (Freeman & Medoff, 1982; Card, 2009). The authors of primary studies also capture industry differences, variations in age cohorts, and capital stock.

Estimation techniques We control for models that are *dynamic*, thus account for the fact that the elasticity may change in response to shocks (estimated in models such as vector autoregression, partial adjustment model, or distributed lag model, among others). We codify studies that account for *unit fixed effects* either using unit dummies or first differences. This method controls for persistent features that could affect the level of skill (or the level of technology used in firms) in specific cohorts of labor force: features such as location (Borjas & Katz, 2007), degree (Kawaguchi & Mori, 2016), age (Angrist, 1995), and industry (Razzak & Timmins, 2008). On the other hand, we also codify studies that account for *time fixed effects*. This method controls for temporal dynamics of unobservable factors changing in time that could affect the skill-biased technical change.

A notorious issue in the empirical literature estimating substitution elasticities is that of potential endogeneity bias. Researchers try to address this problem by instrumenting labor supply, but good instruments are hard to come by. An example of a suitable *instrument* can be found in Ciccone & Peri (2005), who use state and year specific compulsory school attendance and child labor laws as instruments for relative labor supply of more educated workers. Such an approach also corrects for potential attenuation bias resulting from measurement error. We also control for studies using *natural experiments*, which we define as studies having access to arguably exogenous variation in the relative supply of skill. The most prominent recent example is Carneiro *et al.* (2022), who exploit the construction of new colleges in Norway in the 1970s. Natural experiments tackle endogeneity, but generally with the exception of attenuation bias. In addition, natural experiments typically cover only a short period of time, and their estimates might be biased to zero due to adjustment lags because they might not be able to fully capture the long-run effect of labor supply on prices.

Publication characteristics To account for aspects of quality not captured by the variables introduced above, we employ two additional variables. First, we use the number of citations taken from Google Scholar (variable *Citations*) normalized by the number of years since the first draft of the study appeared in Google Scholar. Second, we use the RePEc recursive discounted impact factor, which is available for journal articles as well as working papers (variable *Impact factor*).

E Diagnostics and Robustness Checks of BMA

Table E1: Diagnostics of the benchmark BMA estimation (UIP and dilution priors)

Mean no. regressors	*Draws*	*Burn-ins*	*Time*	*No. models visited*
15.2681	$3 \cdot 10^5$	$1 \cdot 10^5$	1.79 mins	77,558
Model space	*Visited*	*Top models*	*Corr PMP*	*No. obs.*
$1.7 \cdot 10^7$	46.00%	100%	0.9975	654
Model prior	*g-prior*	*Shrinkage-stats*		
Uniform/12	UIP	Av = 0.9985		

Notes: We employ the combination of unit information prior recommended by (Eicher *et al.*, 2011) and dilution prior suggested by George (2010), which accounts for collinearity.

Figure E1: Benchmark BMA model size and convergence (UIP and dilution priors)

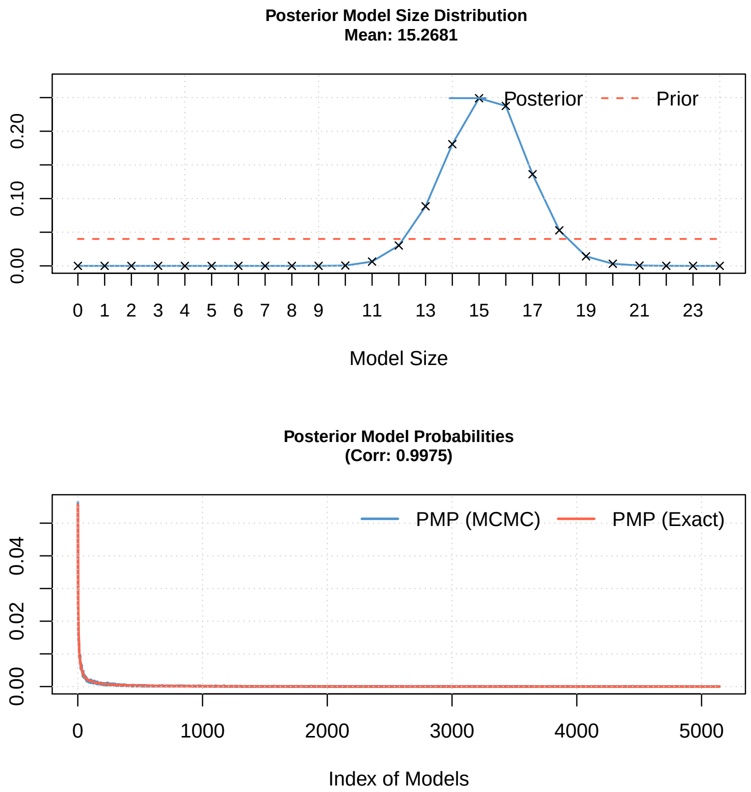
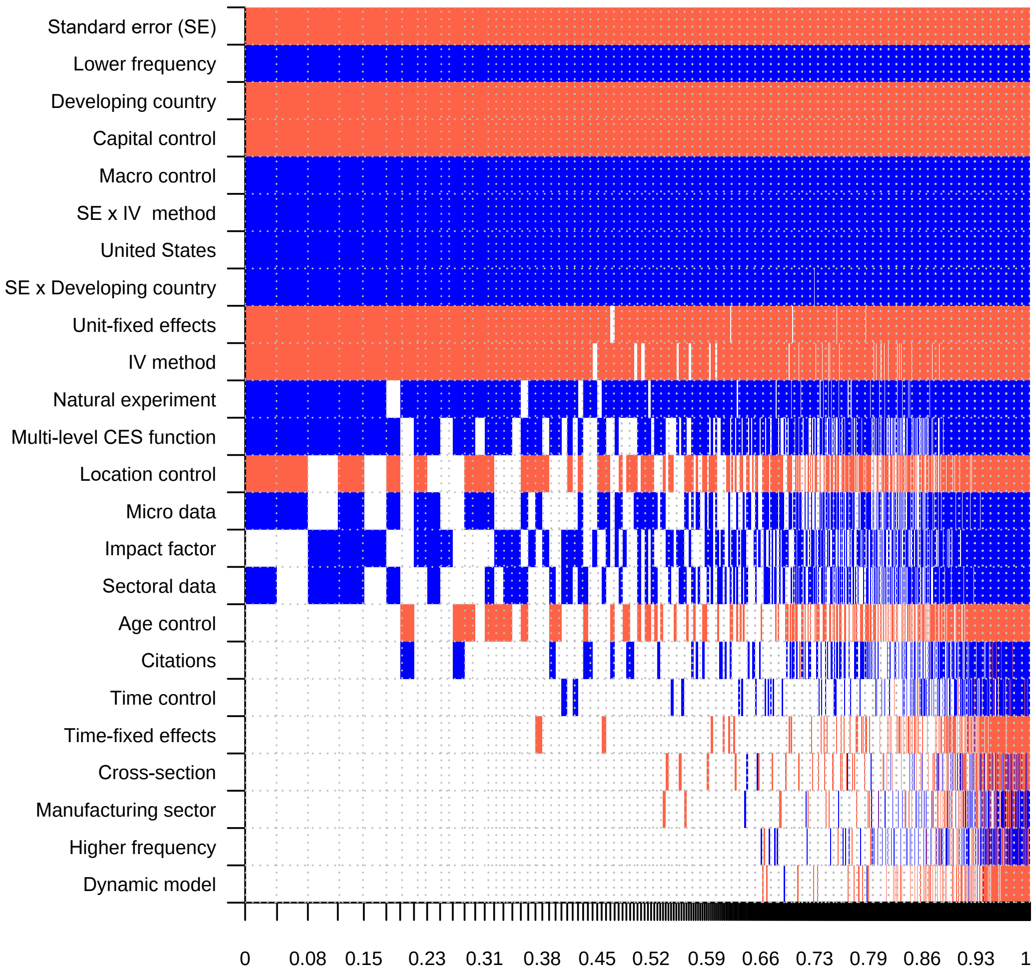


Table E2: Why elasticities vary (alternative priors)

Response variable: Inverse elasticity	Bayesian model averaging (uniform prior)			Bayesian model averaging (BRIC prior)			Bayesian model averaging (hyper-g prior)		
	P. mean	P. SD	PIP	P. mean	P. SD	PIP	P. mean	P. SD	PIP
Constant	-0.19	NA	1.00	-0.20	NA	1.00	-0.21	NA	1.00
Standard error (SE)	-3.62	0.89	1.00	-3.62	0.84	1.00	-3.64	0.80	1.00
SE * IV method	2.31	0.50	1.00	2.35	0.48	1.00	2.35	0.44	1.00
SE * Developing country	2.22	0.61	0.99	2.24	0.59	1.00	2.23	0.56	1.00
Data characteristics									
Higher frequency	0.00	0.01	0.05	0.00	0.02	0.08	-0.01	0.04	0.38
Lower frequency	0.25	0.04	1.00	0.26	0.04	1.00	0.27	0.04	1.00
Micro data	0.05	0.05	0.50	0.06	0.05	0.65	0.08	0.04	0.93
Sectoral data	0.05	0.06	0.46	0.07	0.06	0.61	0.10	0.05	0.91
Cross-section	0.00	0.01	0.06	0.00	0.01	0.10	-0.01	0.03	0.39
Structural variation									
United States	0.10	0.03	1.00	0.10	0.03	1.00	0.10	0.02	1.00
Developing country	-0.21	0.04	1.00	-0.21	0.04	1.00	-0.20	0.04	1.00
Manufacturing sector	0.00	0.02	0.05	0.00	0.02	0.09	-0.01	0.05	0.38
Design of production function									
Multilevel CES function	0.05	0.04	0.67	0.05	0.04	0.79	0.06	0.03	0.95
Time control	0.00	0.01	0.08	0.00	0.01	0.11	0.00	0.02	0.37
Location control	-0.08	0.09	0.53	-0.10	0.08	0.65	-0.13	0.07	0.91
Macro control	0.19	0.04	1.00	0.19	0.04	1.00	0.19	0.03	1.00
Age control	-0.02	0.03	0.31	-0.02	0.03	0.36	-0.02	0.03	0.60
Capital control	-0.39	0.03	1.00	-0.39	0.03	1.00	-0.38	0.03	1.00
Estimation technique									
Dynamic model	0.00	0.01	0.04	0.00	0.02	0.07	-0.01	0.03	0.35
Unit fixed effects	-0.08	0.03	0.97	-0.08	0.02	0.99	-0.08	0.02	1.00
Time fixed effects	0.00	0.01	0.08	0.00	0.01	0.13	-0.01	0.02	0.43
IV method	-0.11	0.05	0.93	-0.12	0.04	0.96	-0.13	0.04	1.00
Natural experiment	0.19	0.09	0.91	0.19	0.08	0.92	0.17	0.07	0.96
Publication characteristics									
Impact factor	0.01	0.01	0.49	0.01	0.01	0.55	0.01	0.01	0.76
Citations	0.00	0.01	0.20	0.00	0.01	0.20	0.00	0.01	0.40
Studies	68			68			68		
Observations	654			654			654		

Notes: P. mean = posterior mean, P. SD = posterior standard deviation, PIP = posterior inclusion probability. In the first specification from the left we employ Bayesian model averaging combining the uniform model prior and the unit information g-prior recommended by Eicher *et al.* (2011). The second specification BRIC and Random = a g-prior by Fernandez *et al.* (2001) for parameters with the beta-binomial model prior (Ley & Steel, 2009) for model space; this ensures that each model size has equal prior probability. The third specification uses a random model prior advocated by Ley & Steel (2009) and the data-dependent hyper-g prior suggested by Feldkircher & Zeugner (2012). All variables are described in Table D1.

Figure E2: Model inclusion in BMA (UIP and uniform priors)



Notes: On the vertical axis the explanatory variables are ranked according to their posterior inclusion probabilities from the highest at the top to the lowest at the bottom. The horizontal axis shows the values of cumulative posterior model probability. Blue color (darker in grayscale) = the estimated parameter of a corresponding explanatory variable is positive. Red color (lighter in grayscale) = the estimated parameter of a corresponding explanatory variable is negative. No color = the corresponding explanatory variable is not included in the model. Numerical results are reported in Table E2. All variables are described in Table D1.

Table E3: Diagnostics of the BMA estimation (UIP and uniform priors)

Mean no. regressors	*Draws*	*Burn-ins*	*Time*	*No. models visited*
14.3781	$3 \cdot 10^5$	$1 \cdot 10^5$	1.45 mins	71,233
Model space	*Visited*	*Top models*	*Corr PMP*	*No. obs.*
$1.7 \cdot 10^7$	42.00%	100%	0.9966	654
Model prior	*g-prior*	*Shrinkage-stats*		
Uniform/12	UIP	Av = 0.9985		

Notes: We employ the priors suggested by Eicher *et al.* (2011), who recommend using the uniform model prior (each model has the same prior probability) and the unit information prior (the prior provides the same amount of information as one observation from the data).

Figure E3: BMA model size and convergence (UIP and uniform priors)

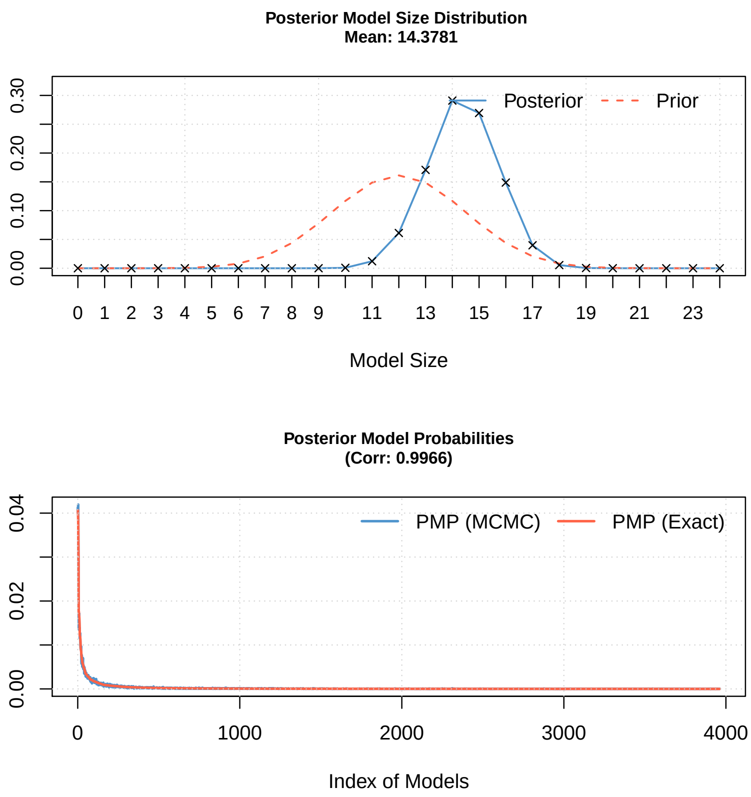
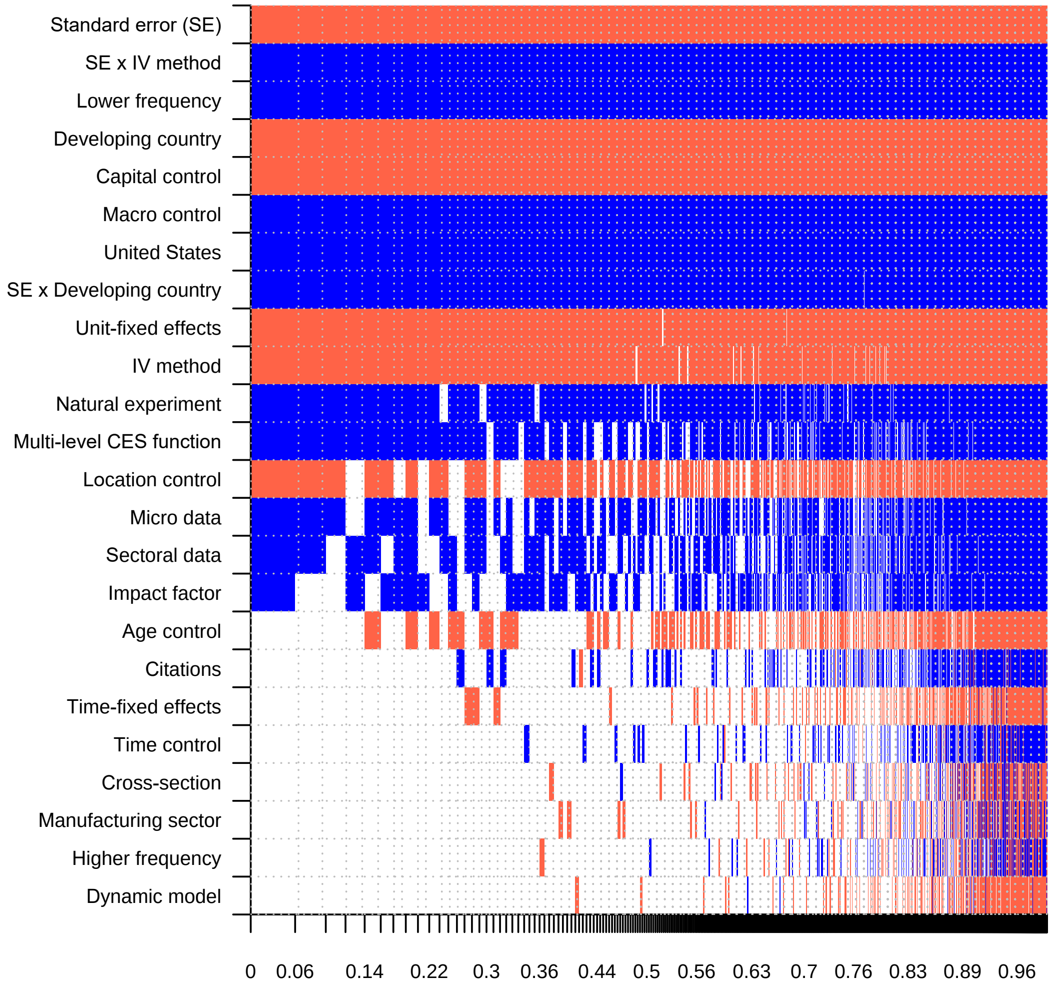


Figure E4: Model inclusion in BMA (BRIC and random priors)



Notes: On the vertical axis the explanatory variables are ranked according to their posterior inclusion probabilities from the highest at the top to the lowest at the bottom. The horizontal axis shows the values of cumulative posterior model probability. Blue color (darker in grayscale) = the estimated parameter of a corresponding explanatory variable is positive. Red color (lighter in grayscale) = the estimated parameter of a corresponding explanatory variable is negative. No color = the corresponding explanatory variable is not included in the model. Numerical results are reported in Table E2. All variables are described in Table D1.

Table E4: Diagnostics of the BMA estimation (BRIC and random priors)

Mean no. regressors	*Draws*	*Burn-ins*	*Time*	*No. models visited*
15.2321	$3 \cdot 10^5$	$1 \cdot 10^5$	1.68 mins	77,762
Model space	*Visited*	*Top models*	*Corr PMP*	*No. obs.*
$1.7 \cdot 10^7$	46.00%	100%	0.9981	654
Model prior	*g-prior*	*Shrinkage-stats*		
Random/12	BRIC	Av = 0.9985		

Notes: The specification uses a BRIC g-prior suggested by Fernandez *et al.* (2001) and the beta-binomial model prior according to Ley & Steel (2009).

Figure E5: BMA model size and convergence (BRIC and random priors)

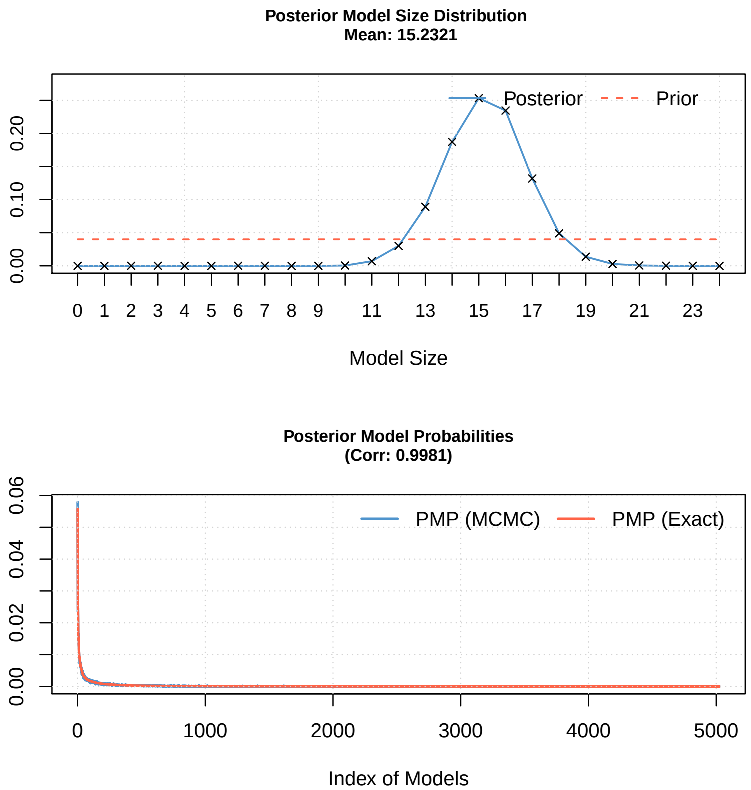
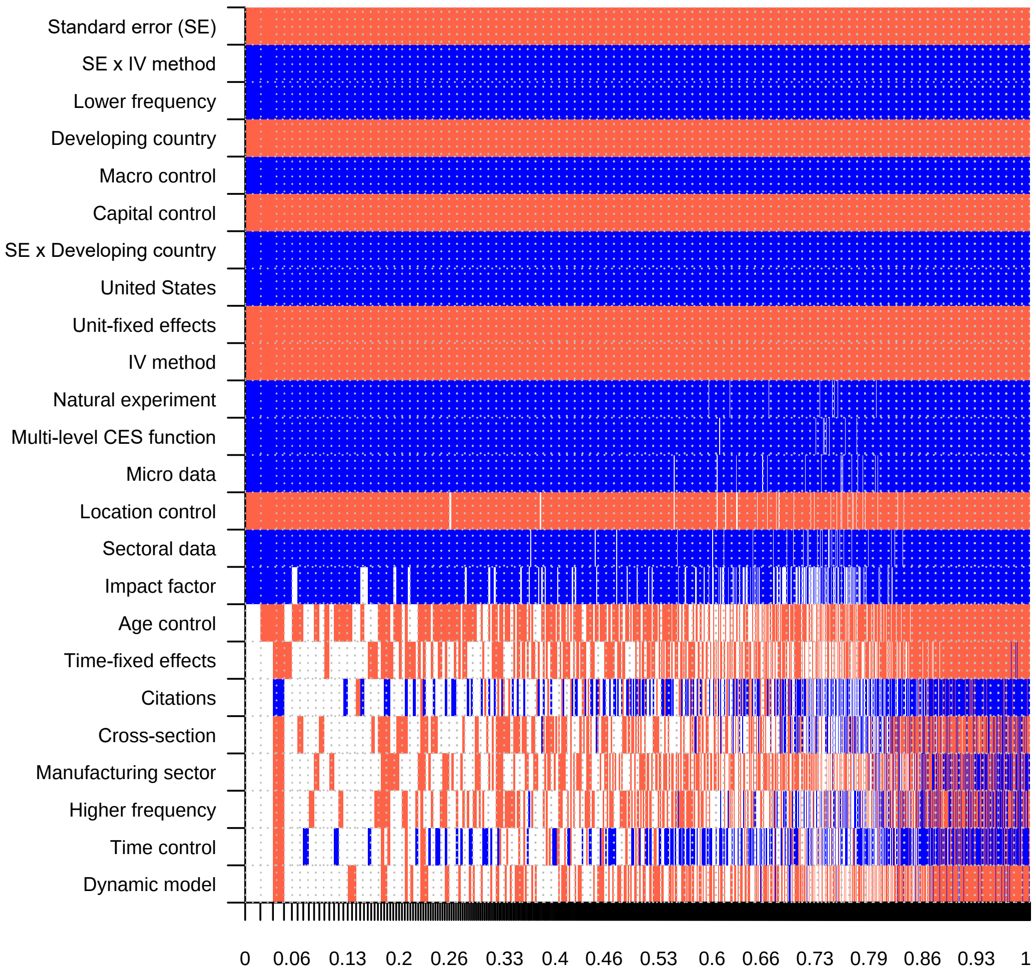


Figure E6: Model inclusion in BMA (hyper-g and random priors)



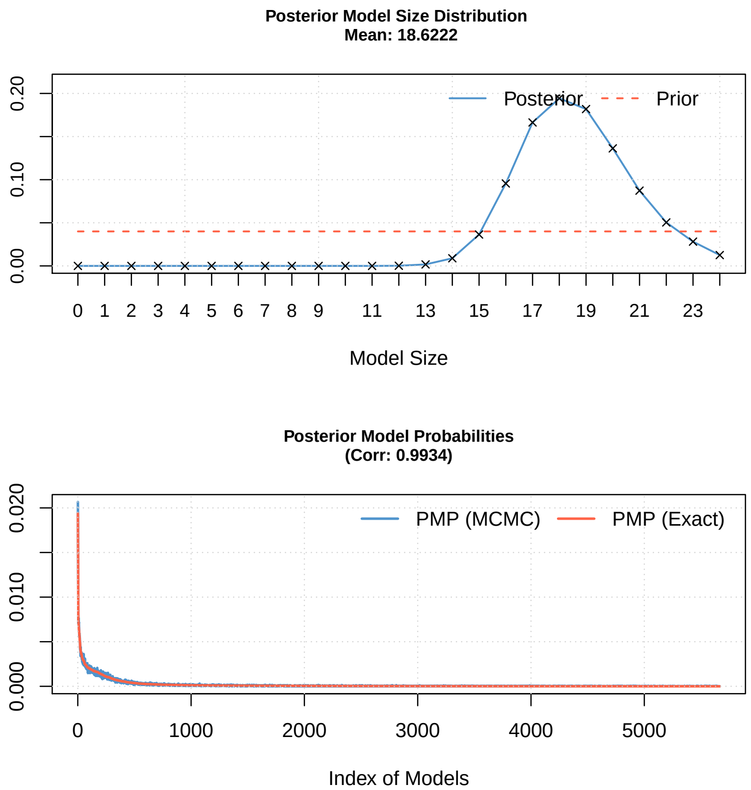
Notes: On the vertical axis the explanatory variables are ranked according to their posterior inclusion probabilities from the highest at the top to the lowest at the bottom. The horizontal axis shows the values of cumulative posterior model probability. Blue color (darker in grayscale) = the estimated parameter of a corresponding explanatory variable is positive. Red color (lighter in grayscale) = the estimated parameter of a corresponding explanatory variable is negative. No color = the corresponding explanatory variable is not included in the model. Numerical results are reported in Table E2. All variables are described in Table D1.

Table E5: Diagnostics of the BMA estimation (hyper-g and random priors)

Mean no. regressors	*Draws*	*Burn-ins*	*Time*	*No. models visited*
18.6222	$3 \cdot 10^5$	$1 \cdot 10^5$	2.52 mins	113,680
Model space	*Visited*	*Top models*	*Corr PMP*	*No. obs.*
$1.7 \cdot 10^7$	68.00%	100%	0.9934	654
Model prior	*g-prior*	*Shrinkage-stats*		
Random/12	hyper (a=2.003058)	Av=0.9814, Stdev=0.0065		

Notes: The specification uses the data-dependent hyper-g prior suggested by Feldkircher & Zeugner (2012) and a random model prior advocated by Ley & Steel (2009).

Figure E7: BMA model size and convergence (hyper-g and random priors)



References

Acemoglu, D. (2002): "Technical change, inequality, and the labor market." *Journal of Economic Literature* **40(1)**: pp. 7–72.

Andrews, I. (2018): "Valid Two-Step Identification-Robust Confidence Sets for GMM." *The Review of Economics and Statistics* **100(2)**: pp. 337–348.

Andrews, I. & M. Kasy (2019): "Identification of and correction for publication bias." *American Economic Review* **109(8)**: pp. 2766–2794.

Angrist, J. D. (1995): "The economic returns to schooling in the West Bank and Gaza Strip." *The American Economic Review* **85(5)**: pp. 1065–1087.

Askilden, J. E. & O. A. Nilsen (2005): "Apprentices and young workers: A study of the Norwegian youth labour market." *Scottish Journal of Political Economy* **52(1)**: pp. 1–17.

Autor, D. H. (2014): "Skills, education, and the rise of earnings inequality among the 'other 99 percent'." *Science* **344(6186)**: pp. 843–851.

Autor, D. H., L. F. Katz, & M. S. Kearney (2008): "Trends in US wage inequality: Revising the revisionists." *The Review of Economics and Statistics* **90(2)**: pp. 300–323.

Avalos, A. & A. Savvides (2006): "The manufacturing wage inequality in Latin America and East Asia: Openness, technology transfer, and labor supply." *Review of Development Economics* **10(4)**: pp. 553–576.

Baum-Snow, N., M. Freedman, & R. Pavan (2018): "Why Has Urban Inequality Increased?" *American Economic Journal: Applied Economics* **10(4)**: pp. 1–42.

Behar, A. (2010): "The elasticity of substitution between skilled and unskilled labor in developing countries is about 2." *Technical report*, Department of Economics, University of Oxford.

Bergstrom, V. & E. E. Panas (1992): "How robust is the capital-skill complementarity hypothesis?" *The Review of Economics and Statistics* **74(3)**: pp. 540–546.

Berman, E., J. Bound, & Z. Griliches (1994): "Changes in the Demand for Skilled Labor within U. S. Manufacturing: Evidence from the Annual Survey of Manufactures." *The Quarterly Journal of Economics* **109(2)**: pp. 367–397.

Berndt, E. R. & L. R. Christensen (1974): "Testing for the existence of a consistent aggregate index of labor inputs." *The American Economic Review* **64(3)**: pp. 391–404.

Berndt, E. R. & C. J. Morrisson (1979): "Income redistribution and employment effects of rising energy prices." *Resources and Energy* **2(2-3)**: pp. 131–150.

Binelli, C. (2015): "How the wage-education profile got more convex: Evidence from Mexico." *The B.E. Journal of Macroeconomics* **15(2)**: pp. 509–560.

Blankenau, W. F. & S. P. Cassou (2011): "Industry estimates of the elasticity of substitution and the rate of biased technological change between skilled and unskilled labour." *Applied Economics* **43(23)**: pp. 3129–3142.

Blundell, R., D. Green, & W. Jin (2016): "The UK wage premium puzzle: How did a large increase in university graduates leave the education premium unchanged?" *IFS Working Papers 1/2016*, The Institute for Fiscal Studies, London.

Boler, E. A. (2016): "Technology-skill complementarity in a globalized world." *Technical report*, Imperial College Business School, Imperial College London, United Kingdom.

Bom, P. R. D. & H. Rachinger (2019): "A kinked meta-regression model for publication bias correction." *Research Synthesis Methods* **10(4)**: pp. 497–514.

Borghans, L. & B. ter Weel (2008): "Understanding the technology of computer technology diffusion: Explaining computer adoption patterns and implications for the wage structure." *Journal of Income Distribution* **17(3-4)**: pp. 37–70.

Borjas, G. J. (2003): "The labor demand curve is downward sloping: Reexamining the impact of immigration on the labor market." *The Quarterly Journal of Economics* **118(4)**: pp. 1335–1374.

Borjas, G. J., J. Grogger, & G. Hanson (2012): "Comment: On Estimating Elasticities Of Substition." *Journal of the European Economic Association* **10(1)**: pp. 198–210.

Borjas, G. J. & L. F. Katz (2007): "The evolution of the Mexican-born workforce in the United States." In G. J. Borjas (editor), "Mexican immigration to the United States," pp. 13–56. National Bureau of Economic Research Conference Report, University of Chicago Press.

Bound, J., J. Groen, G. Kezdi, & S. Turner (2004): "Trade in university training: Cross-state variation in the production and stock of college-educated labor." *Journal of Econometrics* **121(1)**: pp. 143–173.

Bowles, S. (1970): "Aggregation of labor inputs in the economics of growth and planning: Experiments with a two-level CES function." *Journal of Political Economy* **78(1)**: pp. 68–81.

Bowlus, A. J., L. Lochner, C. Robinson, & E. Suleymanoglu (2022): "Wages, skills, and skill-biased technical change: The canonical model revisited." *Journal of Human Resources* **(forthcoming)**.

Broadstock, D., L. Hunt, & S. Sorrell (2007): "UKERC Review of Evidence for the Rebound Effect - Technical report 3: Elasticity of substitution studies." *UKERC Working Paper Series 11/2007*, UK Energy Research Centre, London.

Brucker, H. & E. J. Jahn (2011): "Migration and Wage-Setting: Reassessing the Labor Market Effects of Migration." *Scandinavian Journal of Economics* **113(2)**: pp. 286–317.

Busch, C., D. Krueger, A. Ludwig, I. Popova, &

Z. Iftikhar (2020): “Should Germany have built a new wall? Macroeconomic lessons from the 2015-18 refugee wave.” *Journal of Monetary Economics* **113(C)**: pp. 28–55.

Caliendo, L., L. Opromolla, F. Parro, & A. Sforza (2021): “Goods and factor market integration: A quantitative assessment of the EU enlargement.” *Journal of Political Economy* **129(12)**: pp. 3491–3545.

Card, D. (2009): “Immigration and inequality.” *American Economic Review* **99(2)**: pp. 1–21.

Card, D. & T. Lemieux (2001): “Can falling supply explain the rising return to college for younger men? A cohort-based analysis.” *The Quarterly Journal of Economics* **116(2)**: pp. 705–746.

Carneiro, P., K. Liu, & K. G. Salvanes (2022): “The Supply of Skill and Endogenous Technical Change: Evidence from a College Expansion Reform.” *Journal of European Economic Association* **(forthcoming)**.

Carrasco, R., J. Jimeno, & A. Ortega (2015): “Returns to Skills and the Distribution of Wages: Spain 1995–2010.” *Oxford Bulletin of Economics and Statistics* **77(4)**: pp. 542–565.

Chirinko, R. S. & D. Mallick (2017): “The Substitution Elasticity, Factor Shares, and the Low-Frequency Panel Model.” *American Economic Journal: Macroeconomics* **9(4)**: pp. 225–253.

Choi, K.-S., J.-H. Jeong, & J. Hwa Jung (2005): “The rising supply of college graduates and declining returns for young cohort: The case of Korea.” *Global Economic Review* **34(2)**: pp. 167–180.

Ciccone, A. & G. Peri (2005): “Long-run substitutability between more and less educated workers: Evidence from US states, 1950–1990.” *Review of Economics and Statistics* **87(4)**: pp. 652–663.

Corker, R. & T. Bayoumi (1991): “Apartheid, growth and income distribution in South Africa: Past history and future prospects.” *IMF Working Papers 116/1991*, European Department, International Monetary Fund.

Cruz, M., E. Milet, & M. Olarreaga (2020): “Online exports and the skilled-unskilled wage gap.” *PloS ONE* **15(5)**: p. e0232396.

D’Amuri, F., G. I. Ottaviano, & G. Peri (2010): “The labor market impact of immigration in Western Germany in the 1990s.” *European Economic Review* **54(4)**: pp. 550–570.

Das, G. G. (1999): “What is assumed in the GTAP database’s disaggregation of labor by skill level?” *Monash Economic Working Papers IP-75*, Centre of Policy Studies and Department of Economics, Monash University.

Denny, M. & M. Fuss (1977): “The use of approximation analysis to test for separability and the existence of consistent aggregates.” *The American Economic Review* **67(3)**: pp. 404–418.

Dogan, C. & G. Akay (2019): “The role of labor endowments on industry output in the short run: Evidence from U.S industries.” *International Review of Economics & Finance* **60(C)**: pp. 281–291.

Dougherty, C. R. (1972): “Estimates of labor aggregation functions.” *Journal of Political Economy* **80(6)**: pp. 1101–1119.

Dupuy, A. (2007): “Will the skill-premium in the Netherlands rise in the next decades?” *Applied Economics* **39(21)**: pp. 2723–2731.

Dupuy, A. & P. Marey (2008): “Shifts and twists in the relative productivity of skilled labor.” *Journal of Macroeconomics* **30(2)**: pp. 718–735.

Dustmann, C., J. Ludsteck, & U. Schonberg (2009): “Revisiting the German Wage Structure.” *The Quarterly Journal of Economics* **124(2)**: pp. 843–881.

Eicher, T. S., C. Papageorgiou, & A. E. Raftery (2011): “Default priors and predictive performance in Bayesian model averaging, with application to growth determinants.” *Journal of Applied Econometrics* **26(1)**: pp. 30–55.

Fallon, P. R. & P. R. G. Layard (1975): “Capital-skill complementarity, income distribution, and output accounting.” *Journal of Political Economy* **83(2)**: pp. 279–302.

Feldkircher, M. & S. Zeugner (2012): “The impact of data revisions on the robustness of growth determinants—a note on determinants of economic growth: Will data tell?” *Journal of Applied Econometrics* **27(4)**: pp. 686–694.

Fernandez, C., E. Ley, & M. F. J. Steel (2001): “Benchmark priors for Bayesian Model Averaging.” *Journal of Econometrics* **100(2)**: pp. 381–427.

Fernandez, M. & J. Messina (2018): “Skill premium, labor supply, and changes in the structure of wages in Latin America.” *Journal of Development Economics* **135(C)**: pp. 555–573.

Fitzgerald, J. D. & I. Kearney (2000): “Convergence in living standards in Ireland: The role of the new economy.” *ESRI Working Paper Series 134/2000*, Economic and Social Research Institute.

Foldvari, P. & B. van Leeuwen (2006): “Is there an optimal allocation of skills? A two-level CES approach.” *Technical report*, Department of Economics, University of Warwick.

Freeman, R. B. (1975): “Overinvestment in college training?” *Journal of Human Resources* **10(3)**: pp. 287–311.

Freeman, R. B. & J. L. Medoff (1982): “Substitution between production labor and other Inputs in unionzed and nonunionized manufacturing.” *The Review of Economics and Statistics* **64(2)**: pp. 220–233.

Furukawa, C. (2020): “Publication bias under aggregation frictions: Theory, evidence, and a new correction method.” *Working paper*, MIT.

Gallego, F. A. (2012): “Skill premium in Chile: Studying skill upgrading in the South.” *World De-

velopment **40(3)**: pp. 594–609.

Gancia, G. A., A. Muller, & F. Zilibotti (2013): "Structural development accounting." In D. Acemoglu, M. Arellano, & E. Dekel (editors), "Advances in Economics and Econometrics: Tenth World Congress," volume 2 of *Econometric Society Monographs*, chapter 9, pp. 373–418. Cambridge: Cambridge University Press.

George, E. I. (2010): "Dilution priors: Compensating for model space redundancy." In "IMS Collections Borrowing Strength: Theory Powering Applications – A Festschrift for Lawrence D. Brown," volume 6, p. 158–165. Institute of Mathematical Statistics.

Giannarakis, P. (2017): "Can job displacements explain the UK's productivity puzzle?" *Technical report*, University of Southampton.

Glitz, A. & D. Wissmann (2021): "Skill premiums and the supply of young workers in Germany." *Labour Economics* **72(C)**. Art. 102034.

Goldin, C. D. & L. F. Katz (2009): *The race between education and technology*. Cambridge, MA: Harvard University Press, 1st edition.

Gunadi, C. (2019): "An inquiry on the impact of highly-skilled STEM immigration on the U.S. economy." *Labour Economics* **61(C)**. Art. 101751.

Gyimah-Brempong, K. & A. O. Gyapong (1992): "Elasticities of factor substitution in the production of education." *Economics of Education Review* **11(3)**: pp. 205–217.

Hamermesh, D. S. (1996): *Labor demand*. Princeton, NJ: Princeton University Press.

Havranek, T., T. D. Stanley, H. Doucouliagos, P. R. D. Bom, J. Geyer-Klingeberg, I. Iwasaki, R. W. Reed, K. Rost, & R. C. M. van Aert (2020): "Reporting Guidelines for Meta-Analysis in Economics." *Journal of Economic Surveys* **34(3)**: pp. 469–475.

Heckman, J. J., L. Lochner, & C. Taber (1998): "Explaining rising wage inequality: Explorations with a dynamic general equilibrium model of labor earnings with heterogeneous agents." *Review of Economic Dynamics* **1(1)**: pp. 1–58.

Hendricks, L. & T. Schoellman (2018): "Human Capital and Development Accounting: New Evidence from Wage Gains at Migration." *The Quarterly Journal of Economics* **133(2)**: pp. 665–700.

Hendricks, L. & T. Schoellman (2022): "Skilled Labor Productivity and Cross-country Income Differences." *American Economic Journal: Macroeconomics* **(forthcoming)**.

Hijzen, A., H. Gorg, & R. Hine (2005): "International Outsourcing and the Skill Structure of Labour Demand in the United Kingdom." *The Economic Journal* **115(506)**: pp. 860–878.

Ioannidis, J. P. A., T. Stanley, & H. Doucouliagos (2017): "The Power of Bias in Economics Research." *The Economic Journal* **127(605)**: pp. F236–F265.

Jamet, S. (2005): "From sectoral to macro-economic impact of cuts in social contributions (in French: De l'impact sectoriel a l'effet macro-economique des allegements de cotisations sociales)." *Revue Francaise D'Economie* **19(3)**: pp. 57–90.

Jensen, G. A. & M. A. Morrisey (1986): "The role of physicians in hospital production." *The Review of Economics and Statistics* **68(3)**: pp. 432–442.

Jerzmanowski, M. & R. Tamura (2020): "Aggregate Elasticity of Substitution between Skills: Estimates from a Macroeconomic Approach." *MPRA Paper 100768*, University Library of Munich, Germany.

Johnson, G. E. (1970): "The demand for labor by educational category." *Southern Economic Journal* **37(2)**: pp. 190–204.

Johnson, M. & M. P. Keane (2013): "A Dynamic Equilibrium Model of the US Wage Structure, 1968-1996." *Journal of Labor Economics* **31(1)**: pp. 1–49.

Katz, L. F. & K. M. Murphy (1992): "Changes in relative wages, 1963–1987: Supply and demand factors." *The Quarterly Journal of Economics* **107(1)**: pp. 35–78.

Kawaguchi, D. & Y. Mori (2016): "Why has wage inequality evolved so differently between Japan and the US? The role of the supply of college-educated workers." *Economics of Education Review* **52(1)**: pp. 29–50.

Kearney, I. (1997): "Estimating the demand for skilled labour, unskilled labour and clerical workers: A dynamic framework." *ESRI Working Paper 91*, The Economic and Social Research Institute.

Kesselman, J. R., S. H. Williamson, & E. R. Berndt (1977): "Tax credits for employment rather than investment." *The American Economic Review* **67(3)**: pp. 339–349.

Kim, D. I. (2005): "Growth in college education and wage differentials in Korea." *Seoul Journal of Economics* **18(2)**: pp. 87–123.

Kiyota, K. & Y. Kurokawa (2019): "Intermediate Goods-Skill Complementarity." *Keio-IES Discussion Paper Series 2019-013*, Institute for Economics Studies, Keio University.

Klenow, P. J. & A. Rodriguez-Clare (1997): "The neoclassical revival in growth economics: Has it gone too far?" *NBER Macroeconomics Annual* **12**: pp. 73–114.

Klotz, B., R. Madoo, & R. Hansen (1980): "A study of high and low "labor productivity" establishments in US manufacturing." In J. W. Kendrick & B. N. Vaccara (editors), "New developments in productivity measurement," chapter 3, pp. 239–292. University of Chicago Press.

Kranz, S. & P. Putz (2022): "Methods Matter: p-Hacking and Publication Bias in Causal Analysis in Economics: Comment." *American Economic Review* **(forthcoming)**.

Krusell, P., L. E. Ohanian, J.-V. Rios-Rull, & G. L. Violante (2000): "Capital-skill complementarity and inequality: A macroeconomic analysis." *Econo-*

metrica **68(5)**: pp. 1029–1053.

Kwack, S. Y. (2012): "Wage Inequality and the Contribution of Capital, Differential Labor Quality and Efficiency to Economic Growth in Korea, 1965-2007." *Seoul Journal of Economics* **25(1)**: pp. 1–23.

Ley, E. & M. F. J. Steel (2009): "On the Effect of Prior Assumptions in Bayesian Model Averaging with Applications to Growth Regression." *Journal of Applied Econometrics* **24(4)**: pp. 651–674.

Li, B. (2010): "Multinational production and choice of technologies: New evidence on skill-biased technological change from China." *Economics Letters* **108(2)**: pp. 181–183.

Lindquist, M. J. (2005): "Capital-skill complementarity and inequality in Sweden." *Scandinavian Journal of Economics* **107(4)**: pp. 711–735.

Malmberg, H. (2018): "How does the efficiency of skilled labor vary across rich and poor countries? An analysis using trade and industry data." *Working paper*, Stanford Institute for Economic Policy Research, Stanford University.

Manacorda, M., A. Manning, & J. Wadsworth (2012): "The Impact Of Immigration On The Structure Of Wages: Theory And Evidence From Britain." *Journal of the European Economic Association* **10(1)**: pp. 120–151.

Manacorda, M., C. Sanchez-Paramo, & N. Schady (2010): "Changes in returns to education in Latin America: The role of demand and supply of skills." *Industrial & Labor Relations Review* **63(2)**: pp. 307–326.

McAdam, P. & A. Willman (2018): "Unraveling The Skill Premium." *Macroeconomic Dynamics* **22(1)**: pp. 33–62.

Medina, C. & C. M. Posso (2010): "Technical change and polarization of the labor market: Evidence for Brazil, Colombia and Mexico." *Working Paper Series "Borradores de Economia" 614*, Banco de la Republica de Colombia.

Mello, M. (2011): "Another look at panel estimates of the elasticity of substitution." *Working paper*, IBMEC Business School, Rio de Janeiro.

Mollick, A. V. (2008): "The rise of the skill premium in Mexican maquiladoras." *The Journal of Development Studies* **44(9)**: pp. 1382–1404.

Murphy, K. M., W. C. Riddell, & P. M. Romer (1998): "Wages, skills, and technology in the United States and Canada." *NBER Working Paper 6638*, National Bureau of Economic Research.

Nissim, J. (1984): "The price responsiveness of the demand for labour by skill: British mechanical engineering: 1963-1978." *The Economic Journal* **94(376)**: pp. 812–825.

Ohanian, L., M. Orak, & S. Shen (2021): "Revisiting Capital-Skill Complementarity, Inequality, and Labor Share." *International Finance Discussion Papers 1319*, Board of Governors of the Federal Reserve System (U.S.).

Ottaviano, G. & G. Peri (2012): "Rethinking The Effect Of Immigration On Wages." *Journal of the European Economic Association* **10(1)**: pp. 152–197.

Psacharopoulos, G. & K. Hinchliffe (1972): "Further evidence on the elasticity of substitution among different types of educated labor." *Journal of Political Economy* **80(4)**: pp. 786–792.

Razzak, W. A. & J. C. Timmins (2008): "A macroeconomic perspective on skill shortages and the skill premium in New Zealand." *Australian Economic Papers* **47(1)**: pp. 74–91.

Reijnders, L., M. Timmer, & X. Ye (2021): "Labour demand in global value chains: Is there a bias against unskilled work?" *The World Economy* **44(9)**: pp. 2547–2571.

Reshef, A. (2007): "Heckscher-Ohlin and the global increase of skill premia: Factor intensity reversals to the rescue." *Working paper*, New York University.

Riano, A. (2009): "Trade, technology adoption and the rise of the skill premium in Mexico." *Working paper*, Globalisation and Economic Policy Centre, University of Nottingham.

Robbins, D. J. (1996): "Evidence on trade and wages in the developing world." *Working Papers 119*, Organisation for Economic Co-operation and Development.

Roodman, D., J. G. MacKinnon, M. O. Nielsen, & M. D. Webb (2018): "Fast and wild: Bootstrap inference in Stata using boottest." *Working Paper 1406*, Department of Economics, Queen's University, Canada: Kingston.

Silva, J. C. (2008): "Young employment, job-skill composition and minimum wages: Evidence from a "natural experiment"." *NIPE Working paper 4/2008*, Universidade do Minho, Portugal.

Sun, L. (2018): "Implementing valid two-step identification-robust confidence sets for linear instrumental-variables models." *Stata Journal* **18(4)**: pp. 803–825.

Tinbergen, J. (1974): "Substitution of graduate by other labour." *Kyklos* **27(2)**: pp. 217–226.

te Velde, D. W. & O. Morrissey (2004): "Foreign direct investment, skills and wage inequality in East Asia." *Journal of the Asia Pacific Economy* **9(3)**: pp. 348–369.

Verdugo, G. (2014): "The great compression of the French wage structure, 1969–2008." *Labour Economics* **28(C)**: pp. 131–144.

Wei, X., G. Onel, Z. Guan, & F. M. Roka (2019): "Substitution between immigrant and native farmworkers in the United States: Does legal status matter?" *IZA Journal of Development and Migration* **10(1)**: pp. 1–27.

Welch, F. (1970): "Education in production." *Journal of Political Economy* **78(1)**: pp. 35–59.

Yang, D. (2012): "The responsiveness of firms to labor market changes in China: Evidence from micro level data." *Working Paper 6239*, China Institute for Employment Research and IZA Institute of Labor Economics.